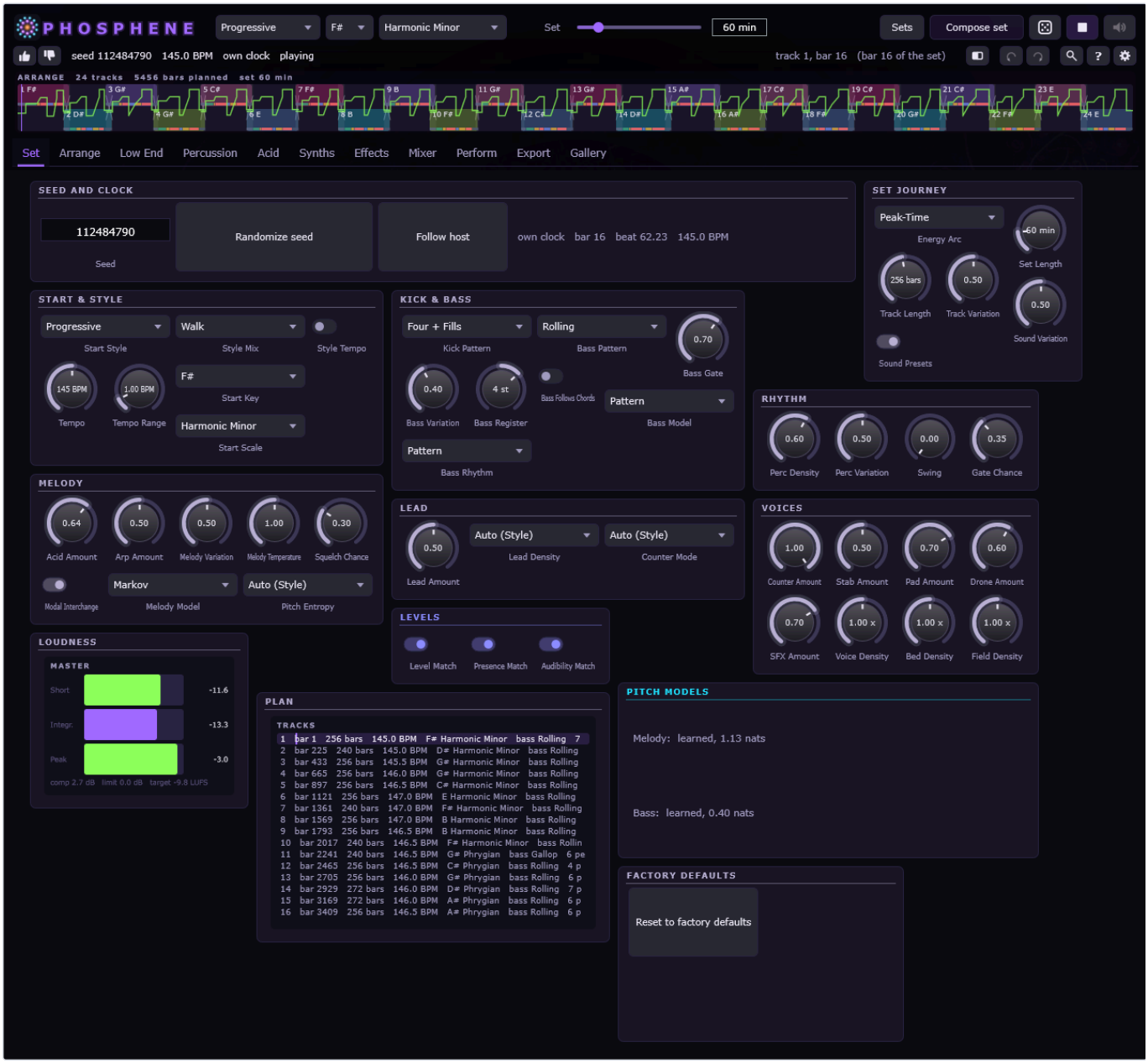
Phosphene

Manual · version 1.6.0



1564 parameters · 18 tabs · 4 perform macros · every picture in this manual is the plugin drawing itself at design size

Contents

1. About
2. Signal flow
3. Set
4. Arrange
5. Kick
6. Bass
7. Percussion
8. Acid
9. Lead
10. Counter
11. Arp
12. Stab
13. Pad
14. Drone
15. SFX / FX
16. Field
17. Mixer / Master
18. Perform
19. Export
20. Gallery
21. Filters and modulation
22. Sound presets
23. The panel
24. Shortcuts
25. Updates

1. About

Phosphene composes and plays complete psytrance sets. You give it a seed, a style and an energy arc; it writes the tracks -- their keys, tempi, forms, patterns and sounds -- and synthesises them in real time. There is no loop pool and no sampled instrument: every note is decided by the composer, and every sound is made by the engine -- except the spoken phrases and, on the Field track, field recordings of real places (and of NASA's missions), which a place cannot be synthesised without.

Two ideas run through the whole instrument. The first is that the same seed always gives the same set: bar 3000 is bar 3000 whether the night was played from the start or joined in the middle, which is what makes a set something you can save in seven lines of text and hand to someone else. The second is that the composer never writes a knob. For every track it chooses a sound preset for each synth and plays it, and on top of that it rides offsets -- a section opening a filter, the level match evening out the parts. The knobs stay yours: the tab shows the preset the composer plays, a thin bright ring shows where a value plays away from its knob, and the moment you turn a knob the synth is yours again (Own Sound).

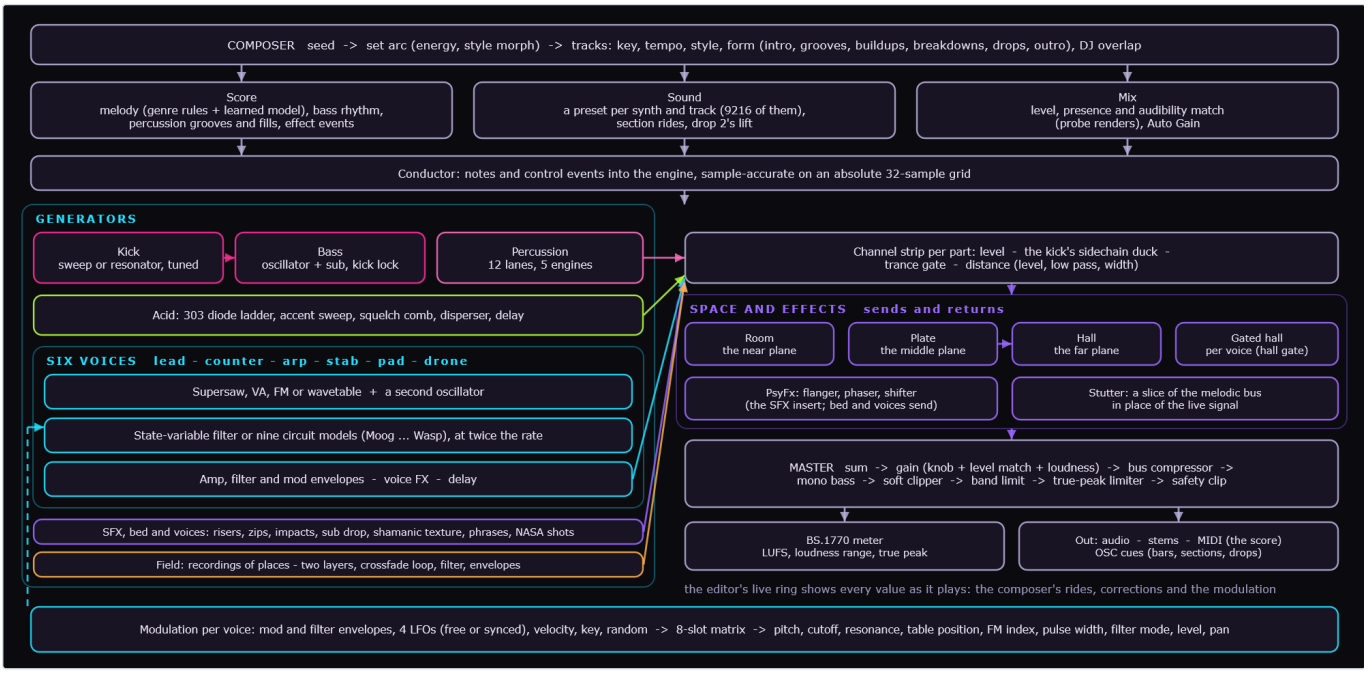
The panel has a tab per generator, plus the pages that are about the set rather than the sound: Set, Arrange, Mixer / Master, Perform and Gallery. Every control on every page is generated from the engine's own parameter tables, and so is this manual.

2. Signal flow

The picture is the whole instrument in the order it works. At the top the composer: a seed becomes a set arc, the arc becomes tracks with keys, tempi, styles and forms, and each track becomes a score, a choice of sounds and a mix -- the last measured by rendering parts of the track before it plays. The conductor hands notes and control events to the engine on an absolute grid of 32 samples, which is why a bar sounds the same however a host cuts its blocks.

On the left the generators, each in the colour of its tab: kick, bass and percussion, the acid, the six melodic voices with their oscillators, filter models and envelopes, the effects with the bed and the spoken voices, and the Field track's sampler. Every part has a channel strip -- level, the kick's sidechain duck, the trance gate and its distance -- and sends into three planes of space: a room near, a plate in the middle, a hall far behind, plus a gated hall a voice can choose. The master sums it all and brings it to loudness: gain, bus compressor, mono bass, soft clipper, band limit, a true-peak limiter and a safety clip, metered to BS.1770. Out go the audio, the stems, the score as MIDI and the cues over OSC.

Along the bottom the modulation of every voice: its envelopes, four LFOs and an eight-slot matrix.



The signal flow, drawn by the plugin (the same picture as the help page's).

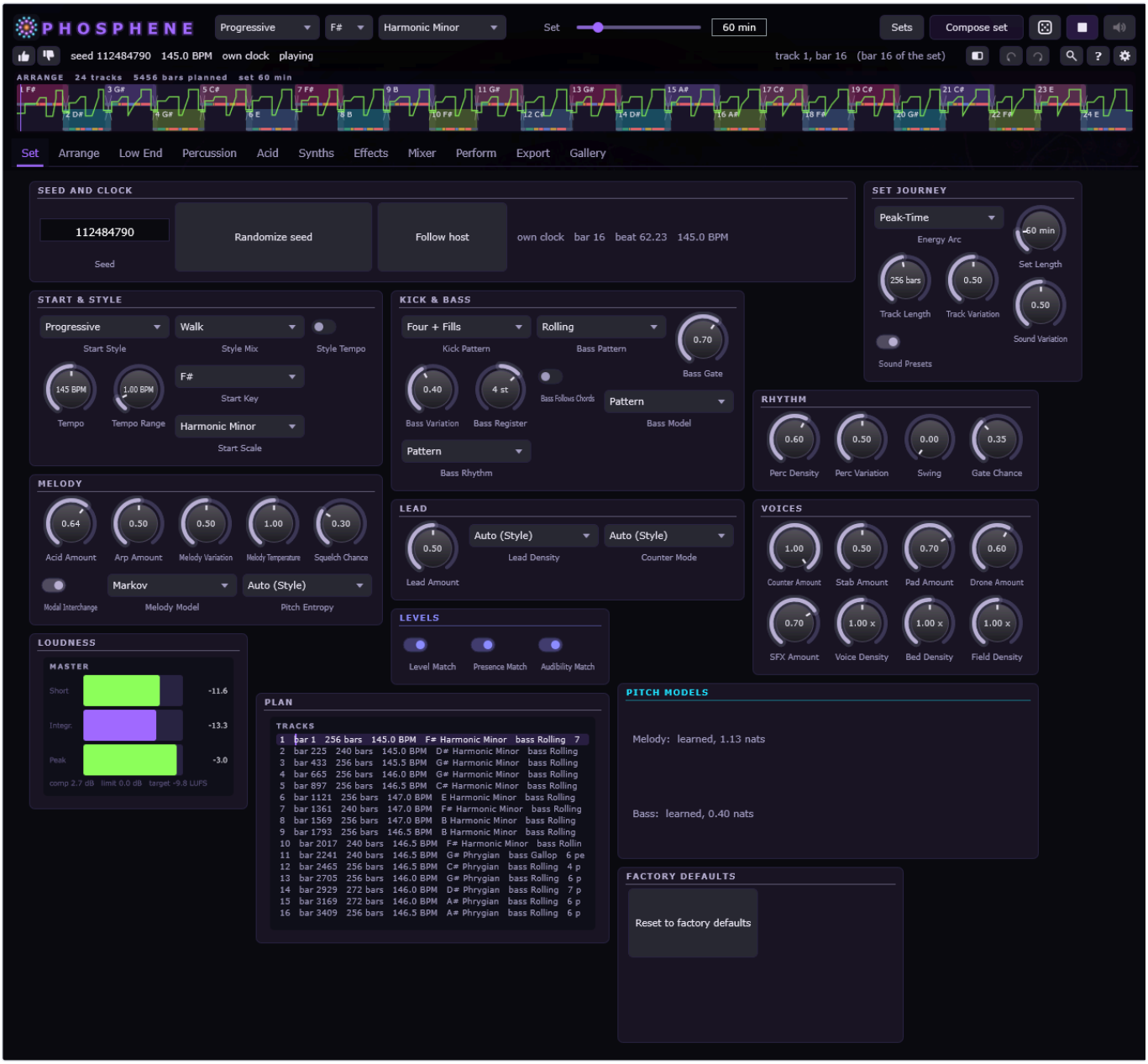
3. Set

The Set tab is where a night begins. The seed names the set; Randomize seed draws another one. Play and Stop run Phosphene's own clock, and Follow host hands the transport to the DAW instead -- under a host the tempo knob follows the host's tempo and the play head follows its position.

Planning a track measures it by rendering parts of it -- its loudness through the master, the brightness of its lines, how audible each line is against the lead. For the first track of a set that happens behind the music: it starts playing at once, its intro runs, and the measured levels ride in over a phrase long before the kick and the lines arrive. A jump to a track that has not been planned yet still waits for its measurements; the header says "planning" rather than going quiet without explanation. In the standalone this has usually happened before you press anything, because the composer thread starts working when the window opens.

The composer's own knobs are here, grouped by what they decide. Start & Style is where the night begins: the style profile, whether the style walks through the set (Style Mix) and whether each track takes its style's own tempo (Style Tempo, which then replaces Tempo and Tempo Range), and the start key and scale. They are called "start" because they are exactly that -- the first track plays them, and from the second on the key, the tempo and (with Style Mix) the style move from track to track as a DJ would move them. Set Journey is the shape of the night: the energy arc, the set length, the track length and how much tracks and sounds vary. Kick & Bass, Rhythm, Melody, Lead and Voices say how much of each part there is and how it is drawn; Levels holds the three automatic balances -- the loudness of whole tracks, the presence of the melodic lines, and the audibility of the quiet lines against the lead.

Export MIDI writes the score of the first N bars as a Standard MIDI File with a named track per part, section markers and the tempo map. The disk on the Export page writes a .phosset -- the seed, the style, the arc, the knobs that differ from their defaults, and every lock and reroll -- which phos_render will play back byte for byte.



The Set tab.

START & STYLE

Control	Key	Range	Default
Start Style	`compose.style`	Goa | Full-On | Progressive | Dark Forest | Hi-Tech	Full-On

Style Mix	`compose.style_mix`	Off | Walk | All	Walk
Style Tempo	`compose.style_tempo`	off / on	off
Tempo	`compose.bpm`	100 .. 190 BPM	145
Tempo Range	`compose.tempo_range`	0 .. 10 BPM	1
Start Key	`compose.key`	C | C# | D | D# | E | F | F# | G | G# | A | A# | B	F#
Start Scale	`compose.scale`	Aeolian | Phrygian | Harmonic Minor | Phrygian Dominant | Double Harmonic | Dorian	Phrygian

SET JOURNEY

Control	Key	Range	Default
Energy Arc	`compose.arc`	Warm-up | Peak-Time | Morning | Closing | Flat	Flat
Set Length	`compose.set_minutes`	10 .. 300 min	60
Track Length	`compose.track_bars`	32 .. 512 bars	256
Track Variation	`compose.track_variation`	0 .. 1	0.5
Sound Variation	`compose.sound_variation`	0 .. 1	0.5
Sound Presets	`compose.sound_presets`	off / on	on

KICK & BASS

Control	Key	Range	Default
Kick Pattern	`compose.kick_pattern`	Four | Four + Fills | Off	Four + Fills
Bass Pattern	`compose.bass_pattern`	Rolling | Gallop | Skip | Offbeat | Triplet	Rolling
Bass Gate	`compose.bass_gate`	0.2 .. 1	0.7
Bass Variation	`compose.bass_variation`	0 .. 1	0.4
Bass Register	`compose.bass_register`	-12 .. 12 st	0
Bass Follows Chords	`compose.bass_follows_chords`	off / on	off
Bass Model	`compose.bass_model`	Pattern | Neural	Pattern
Bass Rhythm	`compose.bass_rhythm`	Pattern | Corpus	Pattern

RHYTHM

Control	Key	Range	Default
Perc Density	`compose.perc_density`	0 .. 1	0.6
Perc Variation	`compose.perc_variation`	0 .. 1	0.5
Swing	`compose.swing`	0 .. 0.3	0
Gate Chance	`compose.gate_chance`	0 .. 1	0.35

MELODY

Control	Key	Range	Default
Acid Amount	`compose.acid_amount`	0 .. 1	0.6
Arp Amount	`compose.arp_amount`	0 .. 1	0.5
Melody Variation	`compose.melody_variation`	0 .. 1	0.5
Melody Temperature	`compose.melody_temperature`	0.5 .. 2	1
Squelch Chance	`compose.squelch_chance`	0 .. 1	0.3
Modal Interchange	`compose.modal_interchange`	off / on	on
Melody Model	`compose.melody_model`	Markov | Neural	Markov
Pitch Entropy	`compose.pitch_entropy`	Auto (Style) | Low | Medium | High	Auto (Style)

LEAD

Control	Key	Range	Default
Lead Amount	`compose.lead_amount`	0 .. 1	0.5
Lead Density	`compose.lead_density`	Auto (Style) | Sparse | Medium | Dense	Auto (Style)
Counter Mode	`compose.counter_mode`	Auto (Style) | Echo | Answer | Timbral | Hocket	Auto (Style)

VOICES

Control	Key	Range	Default
Counter Amount	`compose.counter_amount`	0 .. 1	1
Stab Amount	`compose.stab_amount`	0 .. 1	0.5
Pad Amount	`compose.pad_amount`	0 .. 1	0.7
Drone Amount	`compose.drone_amount`	0 .. 1	0.6
SFX Amount	`compose.sfx_amount`	0 .. 1	0.7

Voice Density	`compose.voice_density`	0 .. 2 x	1
Bed Density	`compose.bed_density`	0 .. 2 x	1
Field Density	`compose.field_density`	0 .. 2 x	1

LEVELS

Control	Key	Range	Default
Level Match	`compose.level_match`	off / on	on
Presence Match	`compose.presence_match`	off / on	on
Audibility Match	`compose.audibility_match`	off / on	on

4. Arrange

The Arrange tab is the whole planned set on one page, and the curation loop that goes with it.

The strip across the top is the night: every planned track a block in the order it plays, the energy arc drawn over it, the play head in it. Underneath, each track gets a row of its own in which its own form fills the full width, so a section stays readable no matter how long the set is. The colours are the section categories: intro and outro grey, groove green, buildup amber, the pre-drop break violet, the drop red, the breakdown blue. Clicking any block jumps there.

The two small glyphs on every track and every section are the answer to the oldest complaint about generators, which is that they are boring. The padlock freezes that unit on the seed it has; the die draws it again. Because a locked unit keeps its old seed and nothing later reads anything an earlier one decided, rerolling one track leaves every other track bit-identical -- so you can keep what you like and reroll what you do not, one section at a time, until the set is yours. Reroll the set draws the whole journey again and leaves the locked tracks exactly where they were.



The Arrange tab.

5. Kick

The kick is the floor of the genre and the first thing to get right. Two engines: a swept sine whose pitch falls from Pitch Start to Pitch End with its own decay, and a damped phasor resonator with the retrigger behaviour of a TR-808. Punch is a second, much faster pitch envelope on top of the first; Click adds the attack transient; Drive and Clip decide how much of the body survives.

Two of its numbers are not free. The tail is limited so that the kick has fallen far enough by the time the first bass note starts, and the phase lock (on the Bass tab) can trim the kick's decay so that the kick and the bass fundamental meet in phase. Both are applied in the engine, after every knob and every variation.

The Kick Scope renders one hit with the values that play right now (the knobs plus the track's recipe; stopped, the knobs): its waveform, its pitch sweep on a log axis, its spectrum, and in figures where the sweep starts, the note it lands on (with its cents off it), its length to -60 dB and its peak.



The Kick tab.

SOUND

Control	Key	Range	Default
Kick Own Sound	`mix.kick_own`	off / on	off

PITCH

Control	Key	Range	Default
Engine	`kick.engine`	Sweep | Resonant	Sweep
Tune	`kick.tune`	Free | Key	Key
Pitch Start	`kick.pitch_start`	60 .. 1500 Hz	330
Pitch End	`kick.pitch_end`	30 .. 120 Hz	50
Body Decay	`kick.pitch_decay`	5 .. 150 ms	13

PUNCH & BODY

Control	Key	Range	Default
Punch	`kick.punch`	0 .. 1	0.5
Punch Decay	`kick.punch_decay`	0.5 .. 20 ms	4
Decay	`kick.amp_decay`	20 .. 1500 ms	240
Attack	`kick.amp_attack`	0 .. 10 ms	0.2
Hold	`kick.amp_hold`	0 .. 150 ms	12

CLICK

Control	Key	Range	Default
Click	`kick.click_level`	0 .. 1	0.5
Click Tone	`kick.click_tone`	500 .. 12000 Hz	4000
Click Decay	`kick.click_decay`	0.5 .. 30 ms	3

DRIVE & OUTPUT

Control	Key	Range	Default
Level	`kick.level`	-36 .. 6 dB	-6
Drive	`kick.drive`	0 .. 1	0.3
Clip	`kick.clip`	Tanh | Hard	Tanh
Tone	`kick.tone`	200 .. 20000 Hz	9000
Tail Limit	`kick.tail_limit`	-60 .. 0 dB	-15

6. Bass

The rolling bass and the kick are one instrument with two halves. The oscillator is a PolyBLEP saw and pulse with a sine sub on the fundamental -- on the fundamental, not an octave down, because the octave-down sub fought the kick -- and the filter is a non-linear zero-delay ladder.

Kick Lock decides which of the two gives way: Bass follows kick sets the bass start phase to the kick's phase at that instant, Kick follows bass trims the kick's decay by bisection until the two agree. Off leaves them where they fall, which at 145 BPM means anything between -157 and +165 degrees.

Pattern and Gate are the note side: which of the five bass patterns the track prefers, and how long a note is as a fraction of its slot. The gate is never allowed to run shorter than half a period of the fundamental.

The bass is a pedal on the root. Its figures at the phrase ends may touch the b2 or the leading tone only as a pickup -- at most two sixteenths, at the end of the bar, into the next downbeat; anywhere else the figure takes the fifth, because under the drone's tonic those notes are a minor ninth in the low register. Bass Follows Chords (Set tab) moves the bass to the root the pad holds, and only where the pad plays; the drone moves with it.

The Bass Scope renders one eighth note of the bass on the key's root with the values that play right now (the knobs plus the track's recipe; stopped, the knobs): its waveform, the filter's cutoff over the note as the synth computes it, its spectrum, and the note and the filter's travel in figures.



The Bass tab.

SOUND

Control	Key	Range	Default
Bass Own Sound	`mix.bass_own`	off / on	off

OSCILLATOR

Control	Key	Range	Default
Wave	`bass.wave`	0 .. 1	0.15
Sub Mode	`bass.sub_mode`	Mixed | Split	Split

Pulse Width	`bass.pulse_width`	0.05 .. 0.95	0.5
Sub	`bass.sub`	0 .. 1	0.3
Sub Octave	`bass.sub_octave`	0 .. 1	0.5
Split	`bass.split_ratio`	1.2 .. 3 x f0	2
Start Phase	`bass.start_phase`	0 .. 1	0.5
Retrigger	`bass.retrigger`	off / on	on
Kick Lock	`bass.kick_lock`	Off | Bass follows kick | Kick follows bass	Kick follows bass

FILTER

Control	Key	Range	Default
Filter Model	`bass.filter_model`	Bass Ladder | Moog Ladder | Prophet (SSM2040) | Juno (IR3109) | Oberheim SEM | Xpander | Diode Ladder | Korg35 (MS-20) | Polivoks | Wasp	Bass Ladder
Cutoff	`bass.cutoff`	20 .. 10000 Hz	240
Resonance	`bass.resonance`	0 .. 1	0.3
Env Amount	`bass.env_amount`	0 .. 8 oct	4
Filter Mode	`bass.filter_mode`	0 .. 1	0
Key Track	`bass.key_track`	0 .. 1	0.6
Vel > Cutoff	`bass.vel_to_cutoff`	0 .. 1	0.25

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	`bass.filt_attack`	0.1 .. 4000 ms	0.1
Filter Decay	`bass.filter_decay`	3 .. 1000 ms	75
Filter Sustain	`bass.filt_sustain`	0 .. 1	0
Filter Release	`bass.filt_release`	5 .. 8000 ms	300

BITE

Control	Key	Range	Default
Bite	`bass.bite`	0 .. 1	0.5
Bite Cutoff	`bass.bite_cutoff`	100 .. 4000 Hz	500
Bite Reso	`bass.bite_resonance`	0 .. 1	0.2
Bite Env	`bass.bite_env`	0 .. 5 oct	3
Bite Decay	`bass.bite_decay`	5 .. 1000 ms	90
Bite Drive	`bass.bite_drive`	0 .. 1	0.5

AMPLIFIER

Control	Key	Range	Default
Drive	`bass.drive`	0 .. 1	0.3
Attack	`bass.amp_attack`	0.1 .. 50 ms	0.8
Decay	`bass.amp_decay`	5 .. 2000 ms	180
Sustain	`bass.amp_sustain`	0 .. 1	0.55
Release	`bass.amp_release`	1 .. 500 ms	10

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	`bass.menv_attack`	0.1 .. 8000 ms	10
Mod Decay	`bass.menv_decay`	5 .. 12000 ms	800
Mod Sustain	`bass.menv_sustain`	0 .. 1	0
Mod Release	`bass.menv_release`	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	`bass.lfo1_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 1 Sync	`bass.lfo1_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 1 Rate	`bass.lfo1_rate`	0.01 .. 40 Hz	1
LFO 1 Retrig	`bass.lfo1_retrig`	off / on	off
LFO 1 Fade	`bass.lfo1_fade`	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	`bass.lfo2_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 2 Sync	`bass.lfo2_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 2 Rate	`bass.lfo2_rate`	0.01 .. 40 Hz	1
LFO 2 Retrig	`bass.lfo2_retrig`	off / on	off
LFO 2 Fade	`bass.lfo2_fade`	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	`bass.lfo3_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 3 Sync	`bass.lfo3_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 3 Rate	`bass.lfo3_rate`	0.01 .. 40 Hz	1
LFO 3 Retrig	`bass.lfo3_retrig`	off / on	off
LFO 3 Fade	`bass.lfo3_fade`	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	`bass.lfo4_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 4 Sync	`bass.lfo4_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 4 Rate	`bass.lfo4_rate`	0.01 .. 40 Hz	1
LFO 4 Retrig	`bass.lfo4_retrig`	off / on	off
LFO 4 Fade	`bass.lfo4_fade`	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	`bass.mx1_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 1 Target	`bass.mx1_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 1 Amount	`bass.mx1_amount`	-1 .. 1	0
Mod 2 Source	`bass.mx2_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 2 Target	`bass.mx2_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 2 Amount	`bass.mx2_amount`	-1 .. 1	0
Mod 3 Source	`bass.mx3_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 3 Target	`bass.mx3_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 3 Amount	`bass.mx3_amount`	-1 .. 1	0
Mod 4 Source	`bass.mx4_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 4 Target	`bass.mx4_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 4 Amount	`bass.mx4_amount`	-1 .. 1	0
Mod 5 Source	`bass.mx5_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 5 Target	`bass.mx5_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 5 Amount	`bass.mx5_amount`	-1 .. 1	0
Mod 6 Source	`bass.mx6_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 6 Target	`bass.mx6_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 6 Amount	`bass.mx6_amount`	-1 .. 1	0
Mod 7 Source	`bass.mx7_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 7 Target	`bass.mx7_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 7 Amount	`bass.mx7_amount`	-1 .. 1	0
Mod 8 Source	`bass.mx8_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 8 Target	`bass.mx8_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 8 Amount	`bass.mx8_amount`	-1 .. 1	0

DUCK & LEVEL

Control	Key	Range	Default
Level	`bass.level`	-36 .. 6 dB	-3
Duck Depth	`bass.duck_depth`	0 .. 1	0.37
Duck Hold	`bass.duck_hold`	0 .. 200 ms	25
Duck Release	`bass.duck_release`	5 .. 500 ms	60

7. Percussion

Twelve lanes, each a complete voice: a tone with its own pitch envelope and FM, four modal sets, the 808 square stack, noise with clap bursts, a filter, a low cut, drive, choke and panning. The role of a lane decides what it plays in the groove and which General MIDI note it exports as.

The patterns are euclidean, rotated to a middle amount of syncopation rather than to none, and the decisions are made per four-bar phrase. Every lane is high-passed at 150 Hz at the least: the bottom of the mix belongs to the kick and the bass, and a tom that reaches into it takes the floor away.

On a metal lane, Noise trades against the square stack: at 1 the squares keep 20 % of their level. Six summed squares are six steady tones, and a cymbal made mostly of them rings like feedback. The default hats and ride therefore run almost all noise, with their low cuts at 3 .. 3.5 kHz.



The Percussion tab.

The 12 lanes of the kit share one table; lane 1 is printed here and the others are the same controls under the keys `perc2.*` to `perc12.*`.

LANE

Control	Key	Range	Default
Active	`perc1.active`	off / on	on
Role	`perc1.role`	Closed Hat | Open Hat | Ride | Crash | Clap | Snare | Rim | Shaker | Tom | Conga | Zap | Blip	Closed Hat
Engine	`perc1.engine`	Noise | Metal | Modal | Tone | FM	Metal
Density	`perc1.density`	0 .. 1	0.6
Choke Group	`perc1.choke`	0 .. 4	1
Cut Tracks Pitch	`perc1.cut_track`	0 .. 2	0

SOURCE

Control	Key	Range	Default
Pitch	`perc1.pitch`	40 .. 12000 Hz	400
Pitch Amount	`perc1.pitch_amount`	1 .. 16 x	1
Pitch Decay	`perc1.pitch_decay`	0.5 .. 300 ms	10
Tune to Key	`perc1.tune`	off / on	off
Shift	`perc1.shift`	-10 .. 10 ms	0
FM Ratio	`perc1.fm_ratio`	0.25 .. 8	1.41
FM Index	`perc1.fm_index`	0 .. 8	0

Modes	`perc1.mode_set`	Membrane | Bar | Harmonic	Membrane
Mode Damping	`perc1.mode_damp`	0 .. 1	0.5
Metal Scale	`perc1.metal_scale`	0.25 .. 4 x	1

NOISE

Control	Key	Range	Default
Noise	`perc1.noise`	0 .. 1	1
Noise Decay	`perc1.noise_decay`	2 .. 3000 ms	35
Bursts	`perc1.bursts`	1 .. 6	1
Burst Spacing	`perc1.burst_spacing`	2 .. 40 ms	10

FILTER & SHAPE

Control	Key	Range	Default
Filter	`perc1.filter`	Low Pass | Band Pass | High Pass	Low Pass
Cutoff	`perc1.cutoff`	100 .. 18000 Hz	12000
Resonance	`perc1.resonance`	0 .. 1	0.05
Low Cut	`perc1.low_cut`	150 .. 8000 Hz	3000
Decay	`perc1.decay`	2 .. 3000 ms	45
Drive	`perc1.drive`	0 .. 1	0

OUTPUT & MOTION

Control	Key	Range	Default
Level	`perc1.level`	-36 .. 6 dB	1.5
Pan	`perc1.pan`	-1 .. 1	0.45
Pan Depth	`perc1.pan_depth`	0 .. 1	1
Pan Period	`perc1.pan_bars`	0.0625 .. 16 bars	0.1875

8. Acid

A diode ladder after Zavalishin, solved from the inside out, with an accent that charges a sweep capacitor the way the original did -- three accents in a row open further than one. Slide is an exponential glide into the next note, Squelch a cutoff start value plus a comb tuned to the note's period. The tempo delay is in beats, so it follows the set.

In the Phrygian modes the b2 is a degree of the acid's own: the F of an E - F - E riff, on any step and held too. The other colour tones -- the upper note of an augmented second -- stay short neighbours that resolve to the tonic.



The Acid tab.

SOUND

Control	Key	Range	Default
Acid Own Sound	`mix.acid_own`	off / on	off

VOICE

Control	Key	Range	Default
Wave	`acid.wave`	0 .. 1	0
Slide Time	`acid.slide_time`	5 .. 200 ms	70
Amp Decay	`acid.amp_decay`	50 .. 4000 ms	900

FILTER

Control	Key	Range	Default
Cutoff	`acid.cutoff`	80 .. 8000 Hz	900
Resonance	`acid.resonance`	0 .. 1	0.72
Env Amount	`acid.env_amount`	0 .. 6 oct	3.5
Decay	`acid.decay`	30 .. 2000 ms	220
Accent	`acid.accent`	0 .. 1	0.7
Key Track	`acid.key_track`	0 .. 1	0.5

SQUELCH & DRIVE

Control	Key	Range	Default
Squelch	`acid.squelch`	off / on	off
Drive	`acid.drive`	0 .. 1	0.85
Squelch Start	`acid.squelch_start`	2 .. 32 x	12
Squelch Time	`acid.squelch_time`	5 .. 150 ms	28
Low Cut	`acid.low_cut`	150 .. 1000 Hz	250
Comb Mix	`acid.comb_mix`	0 .. 1	0.55
Comb Feedback	`acid.comb_feedback`	0 .. 0.97	0.82
Disperse	`acid.disperse`	0 .. 8 x	0
Disperse Freq	`acid.disperse_freq`	200 .. 8000 Hz	1250

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	`acid.menv_attack`	0.1 .. 8000 ms	10
Mod Decay	`acid.menv_decay`	5 .. 12000 ms	800
Mod Sustain	`acid.menv_sustain`	0 .. 1	0
Mod Release	`acid.menv_release`	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	`acid.lfo1_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 1 Sync	`acid.lfo1_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 1 Rate	`acid.lfo1_rate`	0.01 .. 40 Hz	1
LFO 1 Retrig	`acid.lfo1_retrig`	off / on	off
LFO 1 Fade	`acid.lfo1_fade`	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	`acid.lfo2_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 2 Sync	`acid.lfo2_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 2 Rate	`acid.lfo2_rate`	0.01 .. 40 Hz	1
LFO 2 Retrig	`acid.lfo2_retrig`	off / on	off
LFO 2 Fade	`acid.lfo2_fade`	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	`acid.lfo3_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 3 Sync	`acid.lfo3_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 3 Rate	`acid.lfo3_rate`	0.01 .. 40 Hz	1
LFO 3 Retrig	`acid.lfo3_retrig`	off / on	off
LFO 3 Fade	`acid.lfo3_fade`	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	`acid.lfo4_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 4 Sync	`acid.lfo4_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 4 Rate	`acid.lfo4_rate`	0.01 .. 40 Hz	1
LFO 4 Retrig	`acid.lfo4_retrig`	off / on	off

LFO 4 Fade	acid.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	acid.mx1_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 1 Target	acid.mx1_dst	Off | Pitch | Cutoff | Resonance | Level | Pan	Off
Mod 1 Amount	acid.mx1_amount	-1 .. 1	0
Mod 2 Source	acid.mx2_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 2 Target	acid.mx2_dst	Off | Pitch | Cutoff | Resonance | Level | Pan	Off
Mod 2 Amount	acid.mx2_amount	-1 .. 1	0
Mod 3 Source	acid.mx3_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 3 Target	acid.mx3_dst	Off | Pitch | Cutoff | Resonance | Level | Pan	Off
Mod 3 Amount	acid.mx3_amount	-1 .. 1	0
Mod 4 Source	acid.mx4_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 4 Target	acid.mx4_dst	Off | Pitch | Cutoff | Resonance | Level | Pan	Off
Mod 4 Amount	acid.mx4_amount	-1 .. 1	0
Mod 5 Source	acid.mx5_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 5 Target	acid.mx5_dst	Off | Pitch | Cutoff | Resonance | Level | Pan	Off
Mod 5 Amount	acid.mx5_amount	-1 .. 1	0
Mod 6 Source	acid.mx6_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 6 Target	acid.mx6_dst	Off | Pitch | Cutoff | Resonance | Level | Pan	Off
Mod 6 Amount	acid.mx6_amount	-1 .. 1	0
Mod 7 Source	acid.mx7_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 7 Target	acid.mx7_dst	Off | Pitch | Cutoff | Resonance | Level | Pan	Off
Mod 7 Amount	acid.mx7_amount	-1 .. 1	0
Mod 8 Source	acid.mx8_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 8 Target	acid.mx8_dst	Off | Pitch | Cutoff | Resonance | Level | Pan	Off
Mod 8 Amount	acid.mx8_amount	-1 .. 1	0

DELAY

Control	Key	Range	Default
Delay Send	acid.delay_send	0 .. 1	0.25
Delay Feedback	acid.delay_feedback	0 .. 0.9	0.5
Delay Left	acid.delay_left	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	3/16
Delay Right	acid.delay_right	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	1/4
Delay High Pass	acid.delay_high_pass	150 .. 2000 Hz	400
Delay Low Pass	acid.delay_low_pass	800 .. 16000 Hz	4500

SPACE & OUTPUT

Control	Key	Range	Default
Level	acid.level	-36 .. 6 dB	-2.3
Duck	acid.duck	0 .. 1	0.15
Room Send	acid.room_send	0 .. 1	0.1
Plate Send	acid.plate_send	0 .. 1	0.15
Hall Send	acid.hall_send	0 .. 1	0
Hall Gate	acid.hall_gate	off / on	off

9. Lead

A supersaw after Szabo's measurements of the JP-8000 -- the detune curve, the mix law and the random phase per note -- with a dynamic detune that follows the length of the note. The other three oscillator types are a virtual-analogue saw and pulse, two-operator FM and a wavetable.

Everything but the kick and the bass is high-passed at its own fundamental, which is what keeps eight voices of supersaw out of the bass register.

As in the acid, the Phrygian b2 is one of the lead's degrees, on beats 2 and 4 and off the beat; beats 1 and 3 take a tone of the tonic chord.


PHOSPHENE
Progressive
F#
Harmonic Minor
Set
60 min
Sets
Compose set
seed 112484790 145.0 BPM own clock playing
track 1, bar 16 (bar 16 of the set)
ARRANGE 24 tracks 5456 bars planned set 60 min
Set
Arrange
Low End
Percussion
Acid
Synths
Effects
Mixer
Perform
Export
Gallery
Lead
Counter
Arp
Stab
Pad
Drone
SOUND
Lead Own Sound
Psy Pluck: Twilight Pollen (composer)
Preset
OSCILLATOR
Supersaw
Oscillator
Vocal
Table
0.55
Detune
0.75
Mix
0.00
Wave
0.60
Dynamic Detune
0.50
Pulse Width
0.30
Position
Drift
WAVETABLE MOTION
0.00
Position Env
0.15
Position LFO
Position Decay
LFO Period
FM
2.00
FM Ratio
2.50
FM Index
FM Decay
SECOND OSCILLATOR
Off
Oscillator 2
0.28
Osc 2 Mix
Osc 2 Detune
-1 Oct
Osc 2 Interval
FILTER
State Variable
Filter Model
Low Pass
Filter Type
3.88 oct
Env Amount
Filter Mode
Key Track
HP Floor
HP Track
Disperse
Disperse Freq
1897.21 Hz
Cutoff
0.50
Resonance
FILTER ENVELOPE
Filter Attack
Filter Decay
Filter Sustain
Filter Release
AMPLIFIER
0.55 ms
Attack
317.839 ms
Decay
0.05
Sustain
158.92 ms
Release
Velocity
Glide
MOD ENVELOPE
Mod Attack
Mod Decay
Mod Sustain
Mod Release
LFO 1
Sine
LFO 1 Shape
Free
LFO 1 Sync
1.00 Hz
LFO 1 Rate
LFO 1 Retrig
LFO 1 Fade
LFO 2
Sine
LFO 2 Shape
Free
LFO 2 Sync
1.00 Hz
LFO 2 Rate
LFO 2 Retrig
LFO 2 Fade
LFO 3
Sine
LFO 3 Shape
Free
LFO 3 Sync
1.00 Hz
LFO 3 Rate
LFO 3 Retrig
LFO 3 Fade
LFO 4
Sine
LFO 4 Shape
Free
LFO 4 Sync
1.00 Hz
LFO 4 Rate
LFO 4 Retrig
LFO 4 Fade
MOD MATRIX
Velocity
Mod 1 Source
Cutoff
Mod 1 Target
Mod 1 Amount
Random
Mod 2 Source
Cutoff
Mod 2 Target
Mod 2 Amount
Off
Mod 3 Source
Off
Mod 3 Target
Mod 3 Amount
Off
Mod 4 Source
Off
Mod 4 Target
Mod 4 Amount
Off
Mod 5 Source
Off
Mod 5 Target
Mod 5 Amount
Off
Mod 6 Source
Off
Mod 6 Target
Mod 6 Amount
Off
Mod 7 Source
Off
Mod 7 Target
Mod 7 Amount
Off
Mod 8 Source
Off
Mod 8 Target
Mod 8 Amount
VIBRATO & TREMOLO
8.00 beats
Voice LFO
LFO to Cutoff
LFO to Pitch
LFO to Level
VOICE FX
Phaser
Modulation
0.22
Mod Mix
0.80
Mod Depth
Mod Period
Mod Feedback
TRANCE GATE
Trance Gate
Sixteenths
Gate Pattern
0.85
Gate Depth
0.50
Gate Duty
Gate Attack
Gate Release
Gate Tone
DELAY
0.30
Delay Send
0.40
Delay Feedback
3/16
Delay Left
3/8
Delay Right
Delay High Pass
Delay Low Pass
SPACE & IMAGE
0.00
Distance
0.80
Width
Room Send
Plate Send
Hall Send
Hall Gate
Pan
Slow Movement
OUTPUT
-4.00 dB
Level
0.20
Duck
PATTERN
LEAD bars 13 to 16
this part is silent here

The Lead tab.

SOUND

Control	Key	Range	Default
Lead Own Sound	`mix.lead_own`	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	`lead.osc`	Supersaw | VA | FM | Wavetable
Table	`lead.table`	Classic | Vocal | Glass | PWM | Sync | Formant Saw | WaveEdit Hienharm | WaveEdit Junox_ho | WaveEdit Euclidea | WaveEdit Sohler49 | WaveEdit Vocal_fo | WaveEd
Position	`lead.position`	0 .. 1
Detune	`lead.detune`	0 .. 1
Mix	`lead.mix`	0 .. 1
Wave	`lead.wave`	0 .. 1
Dynamic Detune	`lead.dynamic_detune`	0 .. 1
Pulse Width	`lead.pulse_width`	0.05 .. 0.95
Drift	`lead.drift`	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	`lead.pos_env`	-1 .. 1	0
Position LFO	`lead.pos_lfo_depth`	0 .. 0.5	0.15
Position Decay	`lead.pos_decay`	10 .. 8000 ms	1500
LFO Period	`lead.pos_lfo_beats`	0.25 .. 64 beats	16

FM

Control	Key	Range	Default
FM Ratio	`lead.fm_ratio`	0.5 .. 8	2
FM Index	`lead.fm_index`	0 .. 10	2.5
FM Decay	`lead.fm_decay`	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	`lead.osc2`	Off | Supersaw | VA | FM | Wavetable	Off
Osc 2 Mix	`lead.osc2_mix`	0 .. 1	0.28
Osc 2 Interval	`lead.osc2_interval`	-2 Oct | -1 Oct | -5th | Unison | +5th | +1 Oct	-1 Oct
Osc 2 Detune	`lead.osc2_detune`	-50 .. 50 ct	7

FILTER

Control	Key	Range	Default
Filter Model	`lead.filter_model`	State Variable | Moog Ladder | Prophet (SSM2040) | Juno (IR3109) | Oberheim SEM | Xpander | Diode Ladder | Korg35 (MS-20) | Polivoks | Wasp	State Variable
Filter Type	`lead.filter_type`	Low Pass | Band Pass | High Pass | Notch	Low Pass
Cutoff	`lead.cutoff`	200 .. 18000 Hz	10000
Resonance	`lead.resonance`	0 .. 1	0.15
Env Amount	`lead.env_amount`	0 .. 6 oct	2
Filter Mode	`lead.filter_mode`	0 .. 1	0
Key Track	`lead.key_track`	0 .. 1	0.5
HP Floor	`lead.hp_floor`	40 .. 400 Hz	220
HP Track	`lead.hp_track`	0 .. 1 x f0	1
Disperse	`lead.disperse`	0 .. 8 x	0
Disperse Freq	`lead.disperse_freq`	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	`lead.filt_attack`	0.1 .. 4000 ms	0.1
Filter Decay	`lead.filter_decay`	5 .. 3000 ms	400

Filter Sustain	`lead.filt_sustain`	0 .. 1	0
Filter Release	`lead.filt_release`	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	`lead.amp_attack`	0.3 .. 2000 ms	4
Decay	`lead.amp_decay`	5 .. 4000 ms	500
Sustain	`lead.amp_sustain`	0 .. 1	0.75
Release	`lead.amp_release`	5 .. 4000 ms	180
Velocity	`lead.vel_sens`	0 .. 1	0.4
Glide	`lead.glide`	0 .. 400 ms	45

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	`lead.menv_attack`	0.1 .. 8000 ms	10
Mod Decay	`lead.menv_decay`	5 .. 12000 ms	800
Mod Sustain	`lead.menv_sustain`	0 .. 1	0
Mod Release	`lead.menv_release`	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	`lead.lfo1_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 1 Sync	`lead.lfo1_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 1 Rate	`lead.lfo1_rate`	0.01 .. 40 Hz	1
LFO 1 Retrig	`lead.lfo1_retrig`	off / on	off
LFO 1 Fade	`lead.lfo1_fade`	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	`lead.lfo2_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 2 Sync	`lead.lfo2_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 2 Rate	`lead.lfo2_rate`	0.01 .. 40 Hz	1
LFO 2 Retrig	`lead.lfo2_retrig`	off / on	off
LFO 2 Fade	`lead.lfo2_fade`	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	`lead.lfo3_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 3 Sync	`lead.lfo3_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 3 Rate	`lead.lfo3_rate`	0.01 .. 40 Hz	1
LFO 3 Retrig	`lead.lfo3_retrig`	off / on	off
LFO 3 Fade	`lead.lfo3_fade`	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	`lead.lfo4_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 4 Sync	`lead.lfo4_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 4 Rate	`lead.lfo4_rate`	0.01 .. 40 Hz	1
LFO 4 Retrig	`lead.lfo4_retrig`	off / on	off
LFO 4 Fade	`lead.lfo4_fade`	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	`lead.mx1_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 1 Target	`lead.mx1_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 1 Amount	`lead.mx1_amount`	-1 .. 1	0
Mod 2 Source	`lead.mx2_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 2 Target	`lead.mx2_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off

Mod 2 Amount	lead.mx2_amount	-1 .. 1	0
Mod 3 Source	lead.mx3_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 3 Target	lead.mx3_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 3 Amount	lead.mx3_amount	-1 .. 1	0
Mod 4 Source	lead.mx4_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 4 Target	lead.mx4_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 4 Amount	lead.mx4_amount	-1 .. 1	0
Mod 5 Source	lead.mx5_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 5 Target	lead.mx5_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 5 Amount	lead.mx5_amount	-1 .. 1	0
Mod 6 Source	lead.mx6_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 6 Target	lead.mx6_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 6 Amount	lead.mx6_amount	-1 .. 1	0
Mod 7 Source	lead.mx7_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 7 Target	lead.mx7_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 7 Amount	lead.mx7_amount	-1 .. 1	0
Mod 8 Source	lead.mx8_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 8 Target	lead.mx8_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 8 Amount	lead.mx8_amount	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	lead.lfo_beats	0.25 .. 64 beats	8
LFO to Cutoff	lead.lfo_cutoff	0 .. 4 oct	0.3
LFO to Pitch	lead.lfo_pitch	0 .. 50 ct	0
LFO to Level	lead.lfo_amp	0 .. 1	0.06

VOICE FX

Control	Key	Range	Default
Modulation	lead.mod	Off | Flanger | Phaser | Comb	Phaser
Mod Mix	lead.mod_mix	0 .. 1	0.22
Mod Depth	lead.mod_depth	0 .. 1	0.8
Mod Period	lead.mod_beats	0.5 .. 64 beats	16
Mod Feedback	lead.mod_feedback	-0.9 .. 0.9	0.45

TRANCE GATE

Control	Key	Range	Default
Trance Gate	lead.gate	off / on	off
Gate Pattern	lead.gate_pattern	Sixteenths | Eighths | Rolling | Gallop | 3-3-2 | Triplets	Sixteenths
Gate Depth	lead.gate_depth	0 .. 1	0.85
Gate Duty	lead.gate_duty	0.05 .. 1	0.5
Gate Attack	lead.gate_attack	0.5 .. 60 ms	3
Gate Release	lead.gate_release	0.5 .. 120 ms	12
Gate Tone	lead.gate_tone	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	lead.delay_send	0 .. 1	0.3
Delay Feedback	lead.delay_feedback	0 .. 0.9	0.4
Delay Left	lead.delay_left	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	3/16
Delay Right	lead.delay_right	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	3/8
Delay High Pass	lead.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	lead.delay_low_pass	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	lead.distance	0 .. 1	0
Width	lead.width	0 .. 1	0.8

Room Send	`lead.room_send`	0 .. 1	0.1
Plate Send	`lead.plate_send`	0 .. 1	0
Hall Send	`lead.hall_send`	0 .. 1	0
Hall Gate	`lead.hall_gate`	off / on	off
Pan	`lead.pan`	-1 .. 1	-0.2
Slow Movement	`lead.slow_mod`	0 .. 1	0

OUTPUT

Control	Key	Range	Default
Level	`lead.level`	-36 .. 6 dB	-4
Duck	`lead.duck`	0 .. 1	0.2

10. Counter

The counter-lead answers the lead. It plays in the gaps the lead leaves and in a register of its own, so the two never mask each other; Counter Mode on the Set tab (or the style, on Auto) decides whether it echoes, answers or holds against the lead. The engine is the same as the lead's, with its own knobs and its own presets.



The Counter tab.

SOUND

Control	Key	Range	Default
Counter Own Sound	`mix.counter_own`	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	`counter.osc`	Supersaw | VA | FM | Wavetable
Table	`counter.table`	Classic | Vocal | Glass | PWM | Sync | Formant Saw | WaveEdit Hienharm | WaveEdit Junox_ho | WaveEdit Euclidea | WaveEdit Sohler49 | WaveEdit Vocal_fo | Wav
Position	`counter.position`	0 .. 1
Detune	`counter.detune`	0 .. 1
Mix	`counter.mix`	0 .. 1
Wave	`counter.wave`	0 .. 1
Dynamic Detune	`counter.dynamic_detune`	0 .. 1
Pulse Width	`counter.pulse_width`	0.05 .. 0.95
Drift	`counter.drift`	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	`counter.pos_env`	-1 .. 1	0
Position LFO	`counter.pos_lfo_depth`	0 .. 0.5	0.25
Position Decay	`counter.pos_decay`	10 .. 8000 ms	1500
LFO Period	`counter.pos_lfo_beats`	0.25 .. 64 beats	4

FM

Control	Key	Range	Default
FM Ratio	`counter.fm_ratio`	0.5 .. 8	2
FM Index	`counter.fm_index`	0 .. 10	2.5
FM Decay	`counter.fm_decay`	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	`counter.osc2`	Off | Supersaw | VA | FM | Wavetable	Off
Osc 2 Mix	`counter.osc2_mix`	0 .. 1	0.3
Osc 2 Interval	`counter.osc2_interval`	-2 Oct | -1 Oct | -5th | Unison | +5th | +1 Oct	-1 Oct
Osc 2 Detune	`counter.osc2_detune`	-50 .. 50 ct	6

FILTER

Control	Key	Range	Default
Filter Model	`counter.filter_model`	State Variable | Moog Ladder | Prophet (SSM2040) | Juno (IR3109) | Oberheim SEM | Xpander | Diode Ladder | Korg35 (MS-20) | Polivoks | Wasp	State Variable
Filter Type	`counter.filter_type`	Low Pass | Band Pass | High Pass | Notch	Low Pass
Cutoff	`counter.cutoff`	200 .. 18000 Hz	5000
Resonance	`counter.resonance`	0 .. 1	0.2
Env Amount	`counter.env_amount`	0 .. 6 oct	3.2
Filter Mode	`counter.filter_mode`	0 .. 1	0
Key Track	`counter.key_track`	0 .. 1	0.5
HP Floor	`counter.hp_floor`	40 .. 400 Hz	220
HP Track	`counter.hp_track`	0 .. 1 x f0	1
Disperse	`counter.disperse`	0 .. 8 x	0
Disperse Freq	`counter.disperse_freq`	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	`counter.filt_attack`	0.1 .. 4000 ms	0.1

Filter Decay	counter.filter_decay	5 .. 3000 ms	60
Filter Sustain	counter.filt_sustain	0 .. 1	0
Filter Release	counter.filt_release	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	counter.amp_attack	0.3 .. 2000 ms	1
Decay	counter.amp_decay	5 .. 4000 ms	70
Sustain	counter.amp_sustain	0 .. 1	0
Release	counter.amp_release	5 .. 4000 ms	35
Velocity	counter.vel_sens	0 .. 1	0.4
Glide	counter.glide	0 .. 400 ms	0

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	counter.menv_attack	0.1 .. 8000 ms	10
Mod Decay	counter.menv_decay	5 .. 12000 ms	800
Mod Sustain	counter.menv_sustain	0 .. 1	0
Mod Release	counter.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	counter.lfo1_shape	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 1 Sync	counter.lfo1_sync	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 1 Rate	counter.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrig	counter.lfo1_retrig	off / on	off
LFO 1 Fade	counter.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	counter.lfo2_shape	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 2 Sync	counter.lfo2_sync	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 2 Rate	counter.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrig	counter.lfo2_retrig	off / on	off
LFO 2 Fade	counter.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	counter.lfo3_shape	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 3 Sync	counter.lfo3_sync	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 3 Rate	counter.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrig	counter.lfo3_retrig	off / on	off
LFO 3 Fade	counter.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	counter.lfo4_shape	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 4 Sync	counter.lfo4_sync	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 4 Rate	counter.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrig	counter.lfo4_retrig	off / on	off
LFO 4 Fade	counter.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	counter.mx1_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 1 Target	counter.mx1_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 1 Amount	counter.mx1_amount	-1 .. 1	0
Mod 2 Source	counter.mx2_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off

Mod 2 Target	counter.mx2_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 2 Amount	counter.mx2_amount	-1 .. 1	0
Mod 3 Source	counter.mx3_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 3 Target	counter.mx3_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 3 Amount	counter.mx3_amount	-1 .. 1	0
Mod 4 Source	counter.mx4_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 4 Target	counter.mx4_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 4 Amount	counter.mx4_amount	-1 .. 1	0
Mod 5 Source	counter.mx5_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 5 Target	counter.mx5_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 5 Amount	counter.mx5_amount	-1 .. 1	0
Mod 6 Source	counter.mx6_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 6 Target	counter.mx6_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 6 Amount	counter.mx6_amount	-1 .. 1	0
Mod 7 Source	counter.mx7_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 7 Target	counter.mx7_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 7 Amount	counter.mx7_amount	-1 .. 1	0
Mod 8 Source	counter.mx8_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 8 Target	counter.mx8_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 8 Amount	counter.mx8_amount	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	counter.lfo_beats	0.25 .. 64 beats	4
LFO to Cutoff	counter.lfo_cutoff	0 .. 4 oct	0.35
LFO to Pitch	counter.lfo_pitch	0 .. 50 ct	0
LFO to Level	counter.lfo_amp	0 .. 1	0

VOICE FX

Control	Key	Range	Default
Modulation	counter.mod	Off | Flanger | Phaser | Comb	Flanger
Mod Mix	counter.mod_mix	0 .. 1	0.28
Mod Depth	counter.mod_depth	0 .. 1	0.75
Mod Period	counter.mod_beats	0.5 .. 64 beats	4
Mod Feedback	counter.mod_feedback	-0.9 .. 0.9	0.55

TRANCE GATE

Control	Key	Range	Default
Trance Gate	counter.gate	off / on	off
Gate Pattern	counter.gate_pattern	Sixteenths | Eighths | Rolling | Gallop | 3-3-2 | Triplets	Sixteenths
Gate Depth	counter.gate_depth	0 .. 1	0.85
Gate Duty	counter.gate_duty	0.05 .. 1	0.5
Gate Attack	counter.gate_attack	0.5 .. 60 ms	3
Gate Release	counter.gate_release	0.5 .. 120 ms	12
Gate Tone	counter.gate_tone	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	counter.delay_send	0 .. 1	0.4
Delay Feedback	counter.delay_feedback	0 .. 0.9	0.4
Delay Left	counter.delay_left	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	1/16
Delay Right	counter.delay_right	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	1/8
Delay High Pass	counter.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	counter.delay_low_pass	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	counter.distance	0 .. 1	0.5

Width	`counter.width`	0 .. 1	0.7
Room Send	`counter.room_send`	0 .. 1	0
Plate Send	`counter.plate_send`	0 .. 1	0.2
Hall Send	`counter.hall_send`	0 .. 1	0
Hall Gate	`counter.hall_gate`	off / on	off
Pan	`counter.pan`	-1 .. 1	0.2
Slow Movement	`counter.slow_mod`	0 .. 1	0

OUTPUT

Control	Key	Range	Default
Level	`counter.level`	-36 .. 6 dB	-3
Duck	`counter.duck`	0 .. 1	0.2

11. Arp

The same engine as the lead, with its own knobs. The arpeggio runs on chord tones, and when its pitch range would overlap the lead's by more than two semitones it is octaved away: two voices playing in the same two octaves mask each other, and the composer checks that before it writes the notes.

The counter never strikes together with the lead a semitone or a minor ninth apart on beats 1 and 3.


PHOSPHENE
Progressive
F#
Harmonic Minor
Set
60 min
Sets
Compose set
seed 112484790 145.0 BPM own clock playing
track 1, bar 16 (bar 16 of the set)
ARRANGE 24 tracks 5456 bars planned set 60 min
Set
Arrange
Low End
Percussion
Acid
Synths
Effects
Mixer
Perform
Export
Gallery
Lead
Counter
Arp
Stab
Pad
Drone
SOUND
Arp Own Sound
Rolling Sequence: Deep Turbine (composer)
Preset
OSCILLATOR
VA
Oscillator
Vocal
Table
0.30
Detune
0.60
Mix
0.39
Wave
Dynamic Detune
Pulse Width
Drift
0.30
Position
WAVETABLE MOTION
0.00
Position Env
0.15
Position LFO
Position Decay
LFO Period
FM
2.00
FM Ratio
2.50
FM Index
FM Decay
SECOND OSCILLATOR
Off
Oscillator 2
-1 Oct
Osc 2 Interval
0.26
Osc 2 Mix
Osc 2 Detune
FILTER
Prophet (SSM2040)
Filter Model
Low Pass
Filter Type
3.91 oct
Env Amount
Filter Mode
Key Track
HP Floor
HP Track
Disperse
Disperse Freq
700 Hz
Cutoff
0.29
Resonance
FILTER ENVELOPE
Filter Attack
Filter Decay
Filter Sustain
Filter Release
AMPLIFIER
0.67 ms
Attack
100.68 ms
Decay
0.03
Sustain
50.3 ms
Release
Velocity
Glide
MOD ENVELOPE
Mod Attack
Mod Decay
Mod Sustain
Mod Release
LFO 1
Sine
LFO 1 Shape
Free
LFO 1 Sync
1.00 Hz
LFO 1 Rate
LFO 1 Retrig
LFO 1 Fade
LFO 2
Sine
LFO 2 Shape
Free
LFO 2 Sync
1.00 Hz
LFO 2 Rate
LFO 2 Retrig
LFO 2 Fade
LFO 3
Sine
LFO 3 Shape
Free
LFO 3 Sync
1.00 Hz
LFO 3 Rate
LFO 3 Retrig
LFO 3 Fade
LFO 4
Sine
LFO 4 Shape
Free
LFO 4 Sync
1.00 Hz
LFO 4 Rate
LFO 4 Retrig
LFO 4 Fade
MOD MATRIX
Off
Mod 1 Source
Off
Mod 1 Target
Mod 1 Amount
Off
Mod 2 Source
Off
Mod 2 Target
Mod 2 Amount
Off
Mod 3 Source
Off
Mod 3 Target
Mod 3 Amount
Off
Mod 4 Source
Off
Mod 4 Target
Mod 4 Amount
Off
Mod 5 Source
Off
Mod 5 Target
Mod 5 Amount
Off
Mod 6 Source
Off
Mod 6 Target
Mod 6 Amount
Off
Mod 7 Source
Off
Mod 7 Target
Mod 7 Amount
Off
Mod 8 Source
Off
Mod 8 Target
Mod 8 Amount
VIBRATO & TREMOLO
8.00 beats
Voice LFO
LFO to Cutoff
LFO to Pitch
LFO to Level
VOICE FX
Comb
Modulation
-0.20
Mod Mix
0.00
Mod Depth
Mod Period
Mod Feedback
TRANCE GATE
Trance Gate
Sixteenths
Gate Pattern
0.85
Gate Depth
0.50
Gate Duty
Gate Attack
Gate Release
Gate Tone
DELAY
0.35
Delay Send
0.40
Delay Feedback
3/16
Delay Left
1/8
Delay Right
Delay High Pass
Delay Low Pass
SPACE & IMAGE
0.50
Distance
0.60
Width
Room Send
Plate Send
Hall Send
Hall Gate
Pan
Slow Movement
OUTPUT
-12.4 dB
Level
-0.20
Duck
PATTERN
ARP bars 13 to 16
this part is silent here

The Arp tab.

SOUND

Control	Key	Range	Default
Arp Own Sound	`mix.arp_own`	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	`arp.osc`	Supersaw | VA | FM | Wavetable
Table	`arp.table`	Classic | Vocal | Glass | PWM | Sync | Formant Saw | WaveEdit Hienharm | WaveEdit Junox_ho | WaveEdit Euclidea | WaveEdit Sohler49 | WaveEdit Vocal_fo | WaveEdit
Position	`arp.position`	0 .. 1
Detune	`arp.detune`	0 .. 1
Mix	`arp.mix`	0 .. 1
Wave	`arp.wave`	0 .. 1
Dynamic Detune	`arp.dynamic_detune`	0 .. 1
Pulse Width	`arp.pulse_width`	0.05 .. 0.95
Drift	`arp.drift`	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	`arp.pos_env`	-1 .. 1	0
Position LFO	`arp.pos_lfo_depth`	0 .. 0.5	0.15
Position Decay	`arp.pos_decay`	10 .. 8000 ms	1500
LFO Period	`arp.pos_lfo_beats`	0.25 .. 64 beats	16

FM

Control	Key	Range	Default
FM Ratio	`arp.fm_ratio`	0.5 .. 8	2
FM Index	`arp.fm_index`	0 .. 10	2.5
FM Decay	`arp.fm_decay`	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	`arp.osc2`	Off | Supersaw | VA | FM | Wavetable	Off
Osc 2 Mix	`arp.osc2_mix`	0 .. 1	0.26
Osc 2 Interval	`arp.osc2_interval`	-2 Oct | -1 Oct | -5th | Unison | +5th | +1 Oct	-1 Oct
Osc 2 Detune	`arp.osc2_detune`	-50 .. 50 ct	4

FILTER

Control	Key	Range	Default
Filter Model	`arp.filter_model`	State Variable | Moog Ladder | Prophet (SSM2040) | Juno (IR3109) | Oberheim SEM | Xpander | Diode Ladder | Korg35 (MS-20) | Polivoks | Wasp	State Variable
Filter Type	`arp.filter_type`	Low Pass | Band Pass | High Pass | Notch	Low Pass
Cutoff	`arp.cutoff`	200 .. 18000 Hz	3500
Resonance	`arp.resonance`	0 .. 1	0.3
Env Amount	`arp.env_amount`	0 .. 6 oct	2.8
Filter Mode	`arp.filter_mode`	0 .. 1	0
Key Track	`arp.key_track`	0 .. 1	0.5
HP Floor	`arp.hp_floor`	40 .. 400 Hz	220
HP Track	`arp.hp_track`	0 .. 1 x f0	1
Disperse	`arp.disperse`	0 .. 8 x	0
Disperse Freq	`arp.disperse_freq`	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	`arp.filt_attack`	0.1 .. 4000 ms	0.1
Filter Decay	`arp.filter_decay`	5 .. 3000 ms	140
Filter Sustain	`arp.filt_sustain`	0 .. 1	0

Filter Release	`arp.filt_release`	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	`arp.amp_attack`	0.3 .. 2000 ms	0.8
Decay	`arp.amp_decay`	5 .. 4000 ms	220
Sustain	`arp.amp_sustain`	0 .. 1	0
Release	`arp.amp_release`	5 .. 4000 ms	12
Velocity	`arp.vel_sens`	0 .. 1	0.4
Glide	`arp.glide`	0 .. 400 ms	0

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	`arp.menv_attack`	0.1 .. 8000 ms	10
Mod Decay	`arp.menv_decay`	5 .. 12000 ms	800
Mod Sustain	`arp.menv_sustain`	0 .. 1	0
Mod Release	`arp.menv_release`	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	`arp.lfo1_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 1 Sync	`arp.lfo1_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 1 Rate	`arp.lfo1_rate`	0.01 .. 40 Hz	1
LFO 1 Retrig	`arp.lfo1_retrig`	off / on	off
LFO 1 Fade	`arp.lfo1_fade`	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	`arp.lfo2_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 2 Sync	`arp.lfo2_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 2 Rate	`arp.lfo2_rate`	0.01 .. 40 Hz	1
LFO 2 Retrig	`arp.lfo2_retrig`	off / on	off
LFO 2 Fade	`arp.lfo2_fade`	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	`arp.lfo3_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 3 Sync	`arp.lfo3_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 3 Rate	`arp.lfo3_rate`	0.01 .. 40 Hz	1
LFO 3 Retrig	`arp.lfo3_retrig`	off / on	off
LFO 3 Fade	`arp.lfo3_fade`	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	`arp.lfo4_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 4 Sync	`arp.lfo4_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 4 Rate	`arp.lfo4_rate`	0.01 .. 40 Hz	1
LFO 4 Retrig	`arp.lfo4_retrig`	off / on	off
LFO 4 Fade	`arp.lfo4_fade`	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	`arp.mx1_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 1 Target	`arp.mx1_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 1 Amount	`arp.mx1_amount`	-1 .. 1	0
Mod 2 Source	`arp.mx2_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 2 Target	`arp.mx2_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 2 Amount	`arp.mx2_amount`	-1 .. 1	0

Mod 3 Source	arp.mx3_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 3 Target	arp.mx3_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 3 Amount	arp.mx3_amount	-1 .. 1	0
Mod 4 Source	arp.mx4_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 4 Target	arp.mx4_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 4 Amount	arp.mx4_amount	-1 .. 1	0
Mod 5 Source	arp.mx5_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 5 Target	arp.mx5_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 5 Amount	arp.mx5_amount	-1 .. 1	0
Mod 6 Source	arp.mx6_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 6 Target	arp.mx6_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 6 Amount	arp.mx6_amount	-1 .. 1	0
Mod 7 Source	arp.mx7_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 7 Target	arp.mx7_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 7 Amount	arp.mx7_amount	-1 .. 1	0
Mod 8 Source	arp.mx8_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 8 Target	arp.mx8_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 8 Amount	arp.mx8_amount	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	arp.lfo_beats	0.25 .. 64 beats	8
LFO to Cutoff	arp.lfo_cutoff	0 .. 4 oct	0.28
LFO to Pitch	arp.lfo_pitch	0 .. 50 ct	0
LFO to Level	arp.lfo_amp	0 .. 1	0.08

VOICE FX

Control	Key	Range	Default
Modulation	arp.mod	Off | Flanger | Phaser | Comb	Comb
Mod Mix	arp.mod_mix	0 .. 1	0.2
Mod Depth	arp.mod_depth	0 .. 1	0
Mod Period	arp.mod_beats	0.5 .. 64 beats	16
Mod Feedback	arp.mod_feedback	-0.9 .. 0.9	0.6

TRANCE GATE

Control	Key	Range	Default
Trance Gate	arp.gate	off / on	off
Gate Pattern	arp.gate_pattern	Sixteenths | Eighths | Rolling | Gallop | 3-3-2 | Triplets	Sixteenths
Gate Depth	arp.gate_depth	0 .. 1	0.85
Gate Duty	arp.gate_duty	0.05 .. 1	0.5
Gate Attack	arp.gate_attack	0.5 .. 60 ms	3
Gate Release	arp.gate_release	0.5 .. 120 ms	12
Gate Tone	arp.gate_tone	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	arp.delay_send	0 .. 1	0.35
Delay Feedback	arp.delay_feedback	0 .. 0.9	0.4
Delay Left	arp.delay_left	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	3/16
Delay Right	arp.delay_right	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	1/8
Delay High Pass	arp.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	arp.delay_low_pass	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	arp.distance	0 .. 1	0.5
Width	arp.width	0 .. 1	0.6
Room Send	arp.room_send	0 .. 1	0

Plate Send	`arp.plate_send`	0 .. 1	0.2
Hall Send	`arp.hall_send`	0 .. 1	0
Hall Gate	`arp.hall_gate`	off / on	off
Pan	`arp.pan`	-1 .. 1	0
Slow Movement	`arp.slow_mod`	0 .. 1	0

OUTPUT

Control	Key	Range	Default
Level	`arp.level`	-36 .. 6 dB	-5
Duck	`arp.duck`	0 .. 1	0.2

12. Stab

Short chord hits on the off-beats and the phrase ends, the punctuation of a groove. They are voiced from the track's harmony and kept out of the lead's register.


PHOSPHENE
Progressive
F#
Harmonic Minor
Set
60 min
Sets
Compose set
seed 112484790 145.0 BPM own clock playing
track 1, bar 16 (bar 16 of the set)
ARRANGE 24 tracks 5456 bars planned set 60 min
Set
Arrange
Low End
Percussion
Acid
Synths
Effects
Mixer
Perform
Export
Gallery
Lead
Counter
Arp
Stab
Pad
Drone
SOUND
Stab Own Sound
Supersaw Stab: Buried Torch (composer)
Preset
OSCILLATOR
Supersaw
Vocal
Oscillator
Table
0.74
0.83
0.00
0.30
Detune
Mix
Wave
Position
Dynamic Detune
Pulse Width
Drift
WAVETABLE MOTION
0.00
0.15
Position Env
Position LFO
Position Decay
LFO Period
FM
2.00
2.50
FM Ratio
FM Index
FM Decay
SECOND OSCILLATOR
Off
Oscillator 2
-1 Oct
Osc 2 Interval
0.26
Osc 2 Mix
Osc 2 Detune
FILTER
Prophet (SSM2040)
Low Pass
Filter Model
Filter Type
1.93 oct
Env Amount
Filter Mode
Key Track
HP Floor
HP Track
Disperse
Disperse Freq
2500 Hz
Cutoff
0.13
Resonance
FILTER ENVELOPE
Filter Attack
Filter Decay
Filter Sustain
Filter Release
AMPLIFIER
2.84 ms
395.284 ms
0.83
332.574 ms
Attack
Decay
Sustain
Release
Velocity
Glide
MOD ENVELOPE
Mod Attack
Mod Decay
Mod Sustain
Mod Release
LFO 1
Sine
Free
LFO 1 Shape
LFO 1 Sync
0.13 Hz
LFO 1 Rate
LFO 1 Retrig
LFO 1 Fade
LFO 2
Sine
Free
LFO 2 Shape
LFO 2 Sync
1.00 Hz
LFO 2 Rate
LFO 2 Retrig
LFO 2 Fade
LFO 3
Sine
Free
LFO 3 Shape
LFO 3 Sync
1.00 Hz
LFO 3 Rate
LFO 3 Retrig
LFO 3 Fade
LFO 4
Sine
Free
LFO 4 Shape
LFO 4 Sync
1.00 Hz
LFO 4 Rate
LFO 4 Retrig
LFO 4 Fade
MOD MATRIX
LFO 1
Cutoff
Mod 1 Source
Mod 1 Target
Mod 1 Amount
Velocity
Cutoff
Mod 2 Source
Mod 2 Target
Mod 2 Amount
Off
Mod 3 Source
Mod 3 Target
Mod 3 Amount
Off
Off
Mod 4 Source
Mod 4 Target
Mod 4 Amount
Off
Mod 5 Source
Mod 5 Target
Mod 5 Amount
Off
Off
Mod 6 Source
Mod 6 Target
Mod 6 Amount
Off
Mod 7 Source
Mod 7 Target
Mod 7 Amount
Off
Off
Mod 8 Source
Mod 8 Target
Mod 8 Amount
VIBRATO & TREMOLO
8.00 beats
Voice LFO
LFO to Cutoff
LFO to Pitch
LFO to Level
VOICE FX
Off
Modulation
0.00
0.70
Mod Mix
Mod Depth
Mod Period
Mod Feedback
TRANCE GATE
Trance Gate
Sixteenths
Gate Pattern
0.85
0.50
Gate Depth
Gate Duty
Gate Attack
Gate Release
Gate Tone
DELAY
0.45
0.45
3/16
Delay Send
Delay Feedback
Delay Left
1/4
Delay Right
Delay High Pass
Delay Low Pass
SPACE & IMAGE
0.50
0.90
Distance
Width
Room Send
Plate Send
Hall Send
Hall Gate
Pan
Slow Movement
OUTPUT
-7.00 dB
Level
0.20
Duck
PATTERN
STAB bars 17 to 20
this part is silent here

The Stab tab.

SOUND

Control	Key	Range	Default
Stab Own Sound	`mix.stab_own`	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	`stab.osc`	Supersaw | VA | FM | Wavetable
Table	`stab.table`	Classic | Vocal | Glass | PWM | Sync | Formant Saw | WaveEdit Hienharm | WaveEdit Junox_ho | WaveEdit Euclidea | WaveEdit Sohler49 | WaveEdit Vocal_fo | WaveEd
Position	`stab.position`	0 .. 1
Detune	`stab.detune`	0 .. 1
Mix	`stab.mix`	0 .. 1
Wave	`stab.wave`	0 .. 1
Dynamic Detune	`stab.dynamic_detune`	0 .. 1
Pulse Width	`stab.pulse_width`	0.05 .. 0.95
Drift	`stab.drift`	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	`stab.pos_env`	-1 .. 1	0
Position LFO	`stab.pos_lfo_depth`	0 .. 0.5	0.15
Position Decay	`stab.pos_decay`	10 .. 8000 ms	1500
LFO Period	`stab.pos_lfo_beats`	0.25 .. 64 beats	16

FM

Control	Key	Range	Default
FM Ratio	`stab.fm_ratio`	0.5 .. 8	2
FM Index	`stab.fm_index`	0 .. 10	2.5
FM Decay	`stab.fm_decay`	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	`stab.osc2`	Off | Supersaw | VA | FM | Wavetable	Off
Osc 2 Mix	`stab.osc2_mix`	0 .. 1	0.26
Osc 2 Interval	`stab.osc2_interval`	-2 Oct | -1 Oct | -5th | Unison | +5th | +1 Oct	-1 Oct
Osc 2 Detune	`stab.osc2_detune`	-50 .. 50 ct	5

FILTER

Control	Key	Range	Default
Filter Model	`stab.filter_model`	State Variable | Moog Ladder | Prophet (SSM2040) | Juno (IR3109) | Oberheim SEM | Xpander | Diode Ladder | Korg35 (MS-20) | Polivoks | Wasp	State Variable
Filter Type	`stab.filter_type`	Low Pass | Band Pass | High Pass | Notch	Low Pass
Cutoff	`stab.cutoff`	200 .. 18000 Hz	1800
Resonance	`stab.resonance`	0 .. 1	0.35
Env Amount	`stab.env_amount`	0 .. 6 oct	3.5
Filter Mode	`stab.filter_mode`	0 .. 1	0
Key Track	`stab.key_track`	0 .. 1	0.5
HP Floor	`stab.hp_floor`	40 .. 400 Hz	220
HP Track	`stab.hp_track`	0 .. 1 x f0	1
Disperse	`stab.disperse`	0 .. 8 x	0
Disperse Freq	`stab.disperse_freq`	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	`stab.filt_attack`	0.1 .. 4000 ms	0.1
Filter Decay	`stab.filter_decay`	5 .. 3000 ms	90

Filter Sustain	`stab.filt_sustain`	0 .. 1	0
Filter Release	`stab.filt_release`	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	`stab.amp_attack`	0.3 .. 2000 ms	0.5
Decay	`stab.amp_decay`	5 .. 4000 ms	180
Sustain	`stab.amp_sustain`	0 .. 1	0
Release	`stab.amp_release`	5 .. 4000 ms	60
Velocity	`stab.vel_sens`	0 .. 1	0.4
Glide	`stab.glide`	0 .. 400 ms	0

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	`stab.menv_attack`	0.1 .. 8000 ms	10
Mod Decay	`stab.menv_decay`	5 .. 12000 ms	800
Mod Sustain	`stab.menv_sustain`	0 .. 1	0
Mod Release	`stab.menv_release`	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	`stab.lfo1_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 1 Sync	`stab.lfo1_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 1 Rate	`stab.lfo1_rate`	0.01 .. 40 Hz	1
LFO 1 Retrig	`stab.lfo1_retrig`	off / on	off
LFO 1 Fade	`stab.lfo1_fade`	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	`stab.lfo2_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 2 Sync	`stab.lfo2_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 2 Rate	`stab.lfo2_rate`	0.01 .. 40 Hz	1
LFO 2 Retrig	`stab.lfo2_retrig`	off / on	off
LFO 2 Fade	`stab.lfo2_fade`	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	`stab.lfo3_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 3 Sync	`stab.lfo3_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 3 Rate	`stab.lfo3_rate`	0.01 .. 40 Hz	1
LFO 3 Retrig	`stab.lfo3_retrig`	off / on	off
LFO 3 Fade	`stab.lfo3_fade`	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	`stab.lfo4_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 4 Sync	`stab.lfo4_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 4 Rate	`stab.lfo4_rate`	0.01 .. 40 Hz	1
LFO 4 Retrig	`stab.lfo4_retrig`	off / on	off
LFO 4 Fade	`stab.lfo4_fade`	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	`stab.mx1_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 1 Target	`stab.mx1_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 1 Amount	`stab.mx1_amount`	-1 .. 1	0
Mod 2 Source	`stab.mx2_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 2 Target	`stab.mx2_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off

Mod 2 Amount	stab.mx2_amount	-1 .. 1	0
Mod 3 Source	stab.mx3_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 3 Target	stab.mx3_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 3 Amount	stab.mx3_amount	-1 .. 1	0
Mod 4 Source	stab.mx4_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 4 Target	stab.mx4_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 4 Amount	stab.mx4_amount	-1 .. 1	0
Mod 5 Source	stab.mx5_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 5 Target	stab.mx5_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 5 Amount	stab.mx5_amount	-1 .. 1	0
Mod 6 Source	stab.mx6_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 6 Target	stab.mx6_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 6 Amount	stab.mx6_amount	-1 .. 1	0
Mod 7 Source	stab.mx7_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 7 Target	stab.mx7_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 7 Amount	stab.mx7_amount	-1 .. 1	0
Mod 8 Source	stab.mx8_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 8 Target	stab.mx8_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 8 Amount	stab.mx8_amount	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	stab.lfo_beats	0.25 .. 64 beats	8
LFO to Cutoff	stab.lfo_cutoff	0 .. 4 oct	0.22
LFO to Pitch	stab.lfo_pitch	0 .. 50 ct	0
LFO to Level	stab.lfo_amp	0 .. 1	0

VOICE FX

Control	Key	Range	Default
Modulation	stab.mod	Off | Flanger | Phaser | Comb	Off
Mod Mix	stab.mod_mix	0 .. 1	0
Mod Depth	stab.mod_depth	0 .. 1	0.7
Mod Period	stab.mod_beats	0.5 .. 64 beats	16
Mod Feedback	stab.mod_feedback	-0.9 .. 0.9	0.5

TRANCE GATE

Control	Key	Range	Default
Trance Gate	stab.gate	off / on	off
Gate Pattern	stab.gate_pattern	Sixteenths | Eighths | Rolling | Gallop | 3-3-2 | Triplets	Sixteenths
Gate Depth	stab.gate_depth	0 .. 1	0.85
Gate Duty	stab.gate_duty	0.05 .. 1	0.5
Gate Attack	stab.gate_attack	0.5 .. 60 ms	3
Gate Release	stab.gate_release	0.5 .. 120 ms	12
Gate Tone	stab.gate_tone	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	stab.delay_send	0 .. 1	0.45
Delay Feedback	stab.delay_feedback	0 .. 0.9	0.45
Delay Left	stab.delay_left	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	3/16
Delay Right	stab.delay_right	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	1/4
Delay High Pass	stab.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	stab.delay_low_pass	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	stab.distance	0 .. 1	0.5
Width	stab.width	0 .. 1	0.9

Room Send	`stab.room_send`	0 .. 1	0
Plate Send	`stab.plate_send`	0 .. 1	0.15
Hall Send	`stab.hall_send`	0 .. 1	0
Hall Gate	`stab.hall_gate`	off / on	off
Pan	`stab.pan`	-1 .. 1	0
Slow Movement	`stab.slow_mod`	0 .. 1	0

OUTPUT

Control	Key	Range	Default
Level	`stab.level`	-36 .. 6 dB	-7
Duck	`stab.duck`	0 .. 1	0.2

13. Pad

Four-voice voicings with the smallest voice movement between chords, through the wavetable engine. The pads carry the blocks that have no lead and no acid, and the trance gate gives them their movement.

Every pitched voice plays its FM ratio to the nearest half (1, 1.5, 2, ...), so an FM pad or lead is always harmonic: a ratio in between makes the bell and struck-metal spectra that sounded out of tune under a chord. A second oscillator stays within a few cents of its interval, and the pad's partner plays in unison or an octave up, never a fifth, which would stack a second chord on the first.

A pad never holds a minor ninth -- no note a semitone above the tonic that bass and drone play, none a semitone above its own root -- nor the leading tone under the tonic, nor a tritone over its root. Held for bars, those intervals sound wrong rather than dark; the genre makes its dissonance with the sound (clusters, noise, drones), and the Phrygian b2 belongs to the lead and the acid. Where the progression swings to the bII, the pad stays on the tonic and adds the minor sixth instead (E - B - E - C over E). The major seventh is a rare colour, on bVI and bIII only.


PHOSPHENE
Progressive
F#
Harmonic Minor
Set
60 min
Sets
Compose set
seed 112484790 145.0 BPM own clock playing
track 1, bar 16 (bar 16 of the set)
ARRANGE 24 tracks 5456 bars planned set 60 min
Set
Arrange
Low End
Percussion
Acid
Synths
Effects
Mixer
Perform
Export
Gallery
Lead
Counter
Arp
Stab
Pad
Drone
SOUND
Pad Own Sound
Supersaw Pad: Twilight Stupa (composer)
Preset
OSCILLATOR
Supersaw
Vocal
Oscillator
Table
0.85
0.69
0.00
0.30
Detune
Mix
Wave
Position
Dynamic Detune
Pulse Width
Drift
WAVETABLE MOTION
0.30
0.15
Position Env
Position LFO
Position Decay
LFO Period
FM
2.00
2.50
FM Ratio
FM Index
FM Decay
SECOND OSCILLATOR
Off
Oscillator 2
-1 Oct
Osc 2 Interval
0.38
Osc 2 Mix
Osc 2 Detune
FILTER
Juno (IR3109)
Low Pass
Filter Model
Filter Type
2.50 oct
Env Amount
Filter Mode
Key Track
HP Floor
HP Track
Disperse
Disperse Freq
4248.91 Hz
Cutoff
0.24
Resonance
FILTER ENVELOPE
Filter Attack
Filter Decay
Filter Sustain
Filter Release
AMPLIFIER
2.65 ms
776.995 ms
0.74
500 ms
Attack
Decay
Sustain
Release
Velocity
Glide
MOD ENVELOPE
Mod Attack
Mod Decay
Mod Sustain
Mod Release
LFO 1
Sine
Free
LFO 1 Shape
LFO 1 Sync
1.00 Hz
LFO 1 Rate
LFO 1 Retrig
LFO 1 Fade
LFO 2
Sine
Free
LFO 2 Shape
LFO 2 Sync
1.00 Hz
LFO 2 Rate
LFO 2 Retrig
LFO 2 Fade
LFO 3
Sine
Free
LFO 3 Shape
LFO 3 Sync
1.00 Hz
LFO 3 Rate
LFO 3 Retrig
LFO 3 Fade
LFO 4
Sine
Free
LFO 4 Shape
LFO 4 Sync
1.00 Hz
LFO 4 Rate
LFO 4 Retrig
LFO 4 Fade
MOD MATRIX
Off
Mod 1 Source
Mod 1 Target
Mod 1 Amount
Mod 2 Source
Mod 2 Target
Mod 2 Amount
Mod 3 Source
Mod 3 Target
Mod 3 Amount
Mod 4 Source
Mod 4 Target
Mod 4 Amount
Mod 5 Source
Mod 5 Target
Mod 5 Amount
Mod 6 Source
Mod 6 Target
Mod 6 Amount
Mod 7 Source
Mod 7 Target
Mod 7 Amount
Mod 8 Source
Mod 8 Target
Mod 8 Amount
VIBRATO & TREMOLO
16.0 beats
Voice LFO
LFO to Cutoff
LFO to Pitch
LFO to Level
VOICE FX
Off
Modulation
0.00
0.70
Mod Mix
Mod Depth
Mod Period
Mod Feedback
TRANCE GATE
Trance Gate
Sixteenths
Gate Pattern
0.85
Gate Depth
0.50
Gate Duty
Gate Attack
Gate Release
Gate Tone
DELAY
0.00
0.40
Delay Send
Delay Feedback
3/16
Delay Left
1/4
Delay Right
Delay High Pass
Delay Low Pass
SPACE & IMAGE
1.00
1.00
Distance
Width
Room Send
Plate Send
Hall Send
Hall Gate
Pan
Slow Movement
OUTPUT
-16.0 dB
Level
0.50
Duck
PATTERN
PAD bars 17 to 20

The Pad tab.

SOUND

Control	Key	Range	Default
Pad Own Sound	`mix.pad_own`	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	`pad.osc`	Supersaw | VA | FM | Wavetable
Table	`pad.table`	Classic | Vocal | Glass | PWM | Sync | Formant Saw | WaveEdit Hienharm | WaveEdit Junox_ho | WaveEdit Euclidea | WaveEdit Sohler49 | WaveEdit Vocal_fo | WaveEdit
Position	`pad.position`	0 .. 1
Detune	`pad.detune`	0 .. 1
Mix	`pad.mix`	0 .. 1
Wave	`pad.wave`	0 .. 1
Dynamic Detune	`pad.dynamic_detune`	0 .. 1
Pulse Width	`pad.pulse_width`	0.05 .. 0.95
Drift	`pad.drift`	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	`pad.pos_env`	-1 .. 1	0.3
Position LFO	`pad.pos_lfo_depth`	0 .. 0.5	0.15
Position Decay	`pad.pos_decay`	10 .. 8000 ms	3000
LFO Period	`pad.pos_lfo_beats`	0.25 .. 64 beats	16

FM

Control	Key	Range	Default
FM Ratio	`pad.fm_ratio`	0.5 .. 8	2
FM Index	`pad.fm_index`	0 .. 10	2.5
FM Decay	`pad.fm_decay`	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	`pad.osc2`	Off | Supersaw | VA | FM | Wavetable	Off
Osc 2 Mix	`pad.osc2_mix`	0 .. 1	0.38
Osc 2 Interval	`pad.osc2_interval`	-2 Oct | -1 Oct | -5th | Unison | +5th | +1 Oct	-1 Oct
Osc 2 Detune	`pad.osc2_detune`	-50 .. 50 ct	9

FILTER

Control	Key	Range	Default
Filter Model	`pad.filter_model`	State Variable | Moog Ladder | Prophet (SSM2040) | Juno (IR3109) | Oberheim SEM | Xpander | Diode Ladder | Korg35 (MS-20) | Polivoks | Wasp	State Variable
Filter Type	`pad.filter_type`	Low Pass | Band Pass | High Pass | Notch	Low Pass
Cutoff	`pad.cutoff`	200 .. 18000 Hz	5000
Resonance	`pad.resonance`	0 .. 1	0.1
Env Amount	`pad.env_amount`	0 .. 6 oct	0
Filter Mode	`pad.filter_mode`	0 .. 1	0
Key Track	`pad.key_track`	0 .. 1	0.5
HP Floor	`pad.hp_floor`	40 .. 400 Hz	140
HP Track	`pad.hp_track`	0 .. 1 x f0	1
Disperse	`pad.disperse`	0 .. 8 x	0
Disperse Freq	`pad.disperse_freq`	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	`pad.filt_attack`	0.1 .. 4000 ms	0.1
Filter Decay	`pad.filter_decay`	5 .. 3000 ms	400
Filter Sustain	`pad.filt_sustain`	0 .. 1	0

Filter Release	`pad.filt_release`	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	`pad.amp_attack`	0.3 .. 2000 ms	700
Decay	`pad.amp_decay`	5 .. 4000 ms	2000
Sustain	`pad.amp_sustain`	0 .. 1	1
Release	`pad.amp_release`	5 .. 4000 ms	1800
Velocity	`pad.vel_sens`	0 .. 1	0.4
Glide	`pad.glide`	0 .. 400 ms	0

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	`pad.menv_attack`	0.1 .. 8000 ms	10
Mod Decay	`pad.menv_decay`	5 .. 12000 ms	800
Mod Sustain	`pad.menv_sustain`	0 .. 1	0
Mod Release	`pad.menv_release`	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	`pad.lfo1_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 1 Sync	`pad.lfo1_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 1 Rate	`pad.lfo1_rate`	0.01 .. 40 Hz	1
LFO 1 Retrig	`pad.lfo1_retrig`	off / on	off
LFO 1 Fade	`pad.lfo1_fade`	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	`pad.lfo2_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 2 Sync	`pad.lfo2_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 2 Rate	`pad.lfo2_rate`	0.01 .. 40 Hz	1
LFO 2 Retrig	`pad.lfo2_retrig`	off / on	off
LFO 2 Fade	`pad.lfo2_fade`	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	`pad.lfo3_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 3 Sync	`pad.lfo3_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 3 Rate	`pad.lfo3_rate`	0.01 .. 40 Hz	1
LFO 3 Retrig	`pad.lfo3_retrig`	off / on	off
LFO 3 Fade	`pad.lfo3_fade`	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	`pad.lfo4_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 4 Sync	`pad.lfo4_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 4 Rate	`pad.lfo4_rate`	0.01 .. 40 Hz	1
LFO 4 Retrig	`pad.lfo4_retrig`	off / on	off
LFO 4 Fade	`pad.lfo4_fade`	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	`pad.mx1_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 1 Target	`pad.mx1_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 1 Amount	`pad.mx1_amount`	-1 .. 1	0
Mod 2 Source	`pad.mx2_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 2 Target	`pad.mx2_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 2 Amount	`pad.mx2_amount`	-1 .. 1	0

Mod 3 Source	`pad.mx3_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 3 Target	`pad.mx3_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 3 Amount	`pad.mx3_amount`	-1 .. 1	0
Mod 4 Source	`pad.mx4_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 4 Target	`pad.mx4_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 4 Amount	`pad.mx4_amount`	-1 .. 1	0
Mod 5 Source	`pad.mx5_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 5 Target	`pad.mx5_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 5 Amount	`pad.mx5_amount`	-1 .. 1	0
Mod 6 Source	`pad.mx6_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 6 Target	`pad.mx6_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 6 Amount	`pad.mx6_amount`	-1 .. 1	0
Mod 7 Source	`pad.mx7_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 7 Target	`pad.mx7_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 7 Amount	`pad.mx7_amount`	-1 .. 1	0
Mod 8 Source	`pad.mx8_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 8 Target	`pad.mx8_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 8 Amount	`pad.mx8_amount`	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	`pad.lfo_beats`	0.25 .. 64 beats	16
LFO to Cutoff	`pad.lfo_cutoff`	0 .. 4 oct	0.5
LFO to Pitch	`pad.lfo_pitch`	0 .. 50 ct	0
LFO to Level	`pad.lfo_amp`	0 .. 1	0.1

VOICE FX

Control	Key	Range	Default
Modulation	`pad.mod`	Off | Flanger | Phaser | Comb	Off
Mod Mix	`pad.mod_mix`	0 .. 1	0
Mod Depth	`pad.mod_depth`	0 .. 1	0.7
Mod Period	`pad.mod_beats`	0.5 .. 64 beats	16
Mod Feedback	`pad.mod_feedback`	-0.9 .. 0.9	0.5

TRANCE GATE

Control	Key	Range	Default
Trance Gate	`pad.gate`	off / on	off
Gate Pattern	`pad.gate_pattern`	Sixteenths | Eighths | Rolling | Gallop | 3-3-2 | Triplets	Sixteenths
Gate Depth	`pad.gate_depth`	0 .. 1	0.85
Gate Duty	`pad.gate_duty`	0.05 .. 1	0.5
Gate Attack	`pad.gate_attack`	0.5 .. 60 ms	3
Gate Release	`pad.gate_release`	0.5 .. 120 ms	12
Gate Tone	`pad.gate_tone`	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	`pad.delay_send`	0 .. 1	0
Delay Feedback	`pad.delay_feedback`	0 .. 0.9	0.4
Delay Left	`pad.delay_left`	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	3/16
Delay Right	`pad.delay_right`	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	1/4
Delay High Pass	`pad.delay_high_pass`	150 .. 2000 Hz	350
Delay Low Pass	`pad.delay_low_pass`	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	`pad.distance`	0 .. 1	1
Width	`pad.width`	0 .. 1	1
Room Send	`pad.room_send`	0 .. 1	0

Plate Send	`pad.plate_send`	0 .. 1	0.35
Hall Send	`pad.hall_send`	0 .. 1	0
Hall Gate	`pad.hall_gate`	off / on	off
Pan	`pad.pan`	-1 .. 1	0
Slow Movement	`pad.slow_mod`	0 .. 1	0.5

OUTPUT

Control	Key	Range	Default
Level	`pad.level`	-36 .. 6 dB	-16
Duck	`pad.duck`	0 .. 1	0.5

14. Drone

A long, slowly moving tone under the breakdowns and the intros: the sound that holds a track together when the kick is gone. It is high-passed above the bass so it never takes the floor. It holds the tonic, its fifth and its octave, and like the pad never a minor ninth over the bass -- in a DJ blend a note of the incoming track that would stand a semitone over the outgoing tonic waits for the hand-over. With Bass Follows Chords on it follows the bass to the pad's root.



The Drone tab.

SOUND

Control	Key	Range	Default
Drone Own Sound	`mix.drone_own`	off / on	off

OSCILLATOR

Control	Key	Range
Oscillator	`drone.osc`	Supersaw | VA | FM | Wavetable
Table	`drone.table`	Classic | Vocal | Glass | PWM | Sync | Formant Saw | WaveEdit Hienharm | WaveEdit Junox_ho | WaveEdit Euclidea | WaveEdit Sohler49 | WaveEdit Vocal_fo | WaveE
Position	`drone.position`	0 .. 1
Detune	`drone.detune`	0 .. 1
Mix	`drone.mix`	0 .. 1
Wave	`drone.wave`	0 .. 1
Dynamic Detune	`drone.dynamic_detune`	0 .. 1
Pulse Width	`drone.pulse_width`	0.05 .. 0.95
Drift	`drone.drift`	0 .. 8 ct

WAVETABLE MOTION

Control	Key	Range	Default
Position Env	`drone.pos_env`	-1 .. 1	0
Position LFO	`drone.pos_lfo_depth`	0 .. 0.5	0.1
Position Decay	`drone.pos_decay`	10 .. 8000 ms	1500
LFO Period	`drone.pos_lfo_beats`	0.25 .. 64 beats	32

FM

Control	Key	Range	Default
FM Ratio	`drone.fm_ratio`	0.5 .. 8	2
FM Index	`drone.fm_index`	0 .. 10	2.5
FM Decay	`drone.fm_decay`	5 .. 2000 ms	250

SECOND OSCILLATOR

Control	Key	Range	Default
Oscillator 2	`drone.osc2`	Off | Supersaw | VA | FM | Wavetable	Off
Osc 2 Mix	`drone.osc2_mix`	0 .. 1	0.36
Osc 2 Interval	`drone.osc2_interval`	-2 Oct | -1 Oct | -5th | Unison | +5th | +1 Oct	-1 Oct
Osc 2 Detune	`drone.osc2_detune`	-50 .. 50 ct	5

FILTER

Control	Key	Range	Default
Filter Model	`drone.filter_model`	State Variable | Moog Ladder | Prophet (SSM2040) | Juno (IR3109) | Oberheim SEM | Xpander | Diode Ladder | Korg35 (MS-20) | Polivoks | Wasp	State Variable
Filter Type	`drone.filter_type`	Low Pass | Band Pass | High Pass | Notch	Low Pass
Cutoff	`drone.cutoff`	200 .. 18000 Hz	900
Resonance	`drone.resonance`	0 .. 1	0.1
Env Amount	`drone.env_amount`	0 .. 6 oct	0
Filter Mode	`drone.filter_mode`	0 .. 1	0
Key Track	`drone.key_track`	0 .. 1	0.5
HP Floor	`drone.hp_floor`	40 .. 400 Hz	140
HP Track	`drone.hp_track`	0 .. 1 x f0	0.5
Disperse	`drone.disperse`	0 .. 8 x	0
Disperse Freq	`drone.disperse_freq`	200 .. 8000 Hz	1250

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	`drone.filt_attack`	0.1 .. 4000 ms	0.1

Filter Decay	drone.filter_decay	5 .. 3000 ms	400
Filter Sustain	drone.filt_sustain	0 .. 1	0
Filter Release	drone.filt_release	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	drone.amp_attack	0.3 .. 2000 ms	1500
Decay	drone.amp_decay	5 .. 4000 ms	2000
Sustain	drone.amp_sustain	0 .. 1	1
Release	drone.amp_release	5 .. 4000 ms	2500
Velocity	drone.vel_sens	0 .. 1	0.4
Glide	drone.glide	0 .. 400 ms	0

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	drone.menv_attack	0.1 .. 8000 ms	10
Mod Decay	drone.menv_decay	5 .. 12000 ms	800
Mod Sustain	drone.menv_sustain	0 .. 1	0
Mod Release	drone.menv_release	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	drone.lfo1_shape	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 1 Sync	drone.lfo1_sync	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 1 Rate	drone.lfo1_rate	0.01 .. 40 Hz	1
LFO 1 Retrig	drone.lfo1_retrig	off / on	off
LFO 1 Fade	drone.lfo1_fade	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	drone.lfo2_shape	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 2 Sync	drone.lfo2_sync	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 2 Rate	drone.lfo2_rate	0.01 .. 40 Hz	1
LFO 2 Retrig	drone.lfo2_retrig	off / on	off
LFO 2 Fade	drone.lfo2_fade	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	drone.lfo3_shape	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 3 Sync	drone.lfo3_sync	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 3 Rate	drone.lfo3_rate	0.01 .. 40 Hz	1
LFO 3 Retrig	drone.lfo3_retrig	off / on	off
LFO 3 Fade	drone.lfo3_fade	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	drone.lfo4_shape	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 4 Sync	drone.lfo4_sync	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 4 Rate	drone.lfo4_rate	0.01 .. 40 Hz	1
LFO 4 Retrig	drone.lfo4_retrig	off / on	off
LFO 4 Fade	drone.lfo4_fade	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	drone.mx1_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 1 Target	drone.mx1_dst	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 1 Amount	drone.mx1_amount	-1 .. 1	0
Mod 2 Source	drone.mx2_src	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off

Mod 2 Target	`drone.mx2_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 2 Amount	`drone.mx2_amount`	-1 .. 1	0
Mod 3 Source	`drone.mx3_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 3 Target	`drone.mx3_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 3 Amount	`drone.mx3_amount`	-1 .. 1	0
Mod 4 Source	`drone.mx4_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 4 Target	`drone.mx4_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 4 Amount	`drone.mx4_amount`	-1 .. 1	0
Mod 5 Source	`drone.mx5_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 5 Target	`drone.mx5_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 5 Amount	`drone.mx5_amount`	-1 .. 1	0
Mod 6 Source	`drone.mx6_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 6 Target	`drone.mx6_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 6 Amount	`drone.mx6_amount`	-1 .. 1	0
Mod 7 Source	`drone.mx7_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 7 Target	`drone.mx7_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 7 Amount	`drone.mx7_amount`	-1 .. 1	0
Mod 8 Source	`drone.mx8_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 8 Target	`drone.mx8_dst`	Off | Pitch | Osc 2 Pitch | Pulse Width | Table Position | FM Index | Cutoff | Resonance | Filter Mode | Level | Pan	Off
Mod 8 Amount	`drone.mx8_amount`	-1 .. 1	0

VIBRATO & TREMOLO

Control	Key	Range	Default
Voice LFO	`drone.lfo_beats`	0.25 .. 64 beats	32
LFO to Cutoff	`drone.lfo_cutoff`	0 .. 4 oct	0.32
LFO to Pitch	`drone.lfo_pitch`	0 .. 50 ct	0
LFO to Level	`drone.lfo_amp`	0 .. 1	0.08

VOICE FX

Control	Key	Range	Default
Modulation	`drone.mod`	Off | Flanger | Phaser | Comb	Off
Mod Mix	`drone.mod_mix`	0 .. 1	0
Mod Depth	`drone.mod_depth`	0 .. 1	0.7
Mod Period	`drone.mod_beats`	0.5 .. 64 beats	16
Mod Feedback	`drone.mod_feedback`	-0.9 .. 0.9	0.5

TRANCE GATE

Control	Key	Range	Default
Trance Gate	`drone.gate`	off / on	off
Gate Pattern	`drone.gate_pattern`	Sixteenths | Eighths | Rolling | Gallop | 3-3-2 | Triplets	Sixteenths
Gate Depth	`drone.gate_depth`	0 .. 1	0.85
Gate Duty	`drone.gate_duty`	0.05 .. 1	0.5
Gate Attack	`drone.gate_attack`	0.5 .. 60 ms	3
Gate Release	`drone.gate_release`	0.5 .. 120 ms	12
Gate Tone	`drone.gate_tone`	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	`drone.delay_send`	0 .. 1	0
Delay Feedback	`drone.delay_feedback`	0 .. 0.9	0.4
Delay Left	`drone.delay_left`	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	3/16
Delay Right	`drone.delay_right`	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	1/4
Delay High Pass	`drone.delay_high_pass`	150 .. 2000 Hz	350
Delay Low Pass	`drone.delay_low_pass`	800 .. 16000 Hz	6000

SPACE & IMAGE

Control	Key	Range	Default
Distance	`drone.distance`	0 .. 1	1

Width	`drone.width`	0 .. 1	1
Room Send	`drone.room_send`	0 .. 1	0
Plate Send	`drone.plate_send`	0 .. 1	0.5
Hall Send	`drone.hall_send`	0 .. 1	0
Hall Gate	`drone.hall_gate`	off / on	off
Pan	`drone.pan`	-1 .. 1	0
Slow Movement	`drone.slow_mod`	0 .. 1	0.25

OUTPUT

Control	Key	Range	Default
Level	`drone.level`	-36 .. 6 dB	-17
Duck	`drone.duck`	0 .. 1	0.45

15. SFX / FX

The effect generator makes risers, downlifters, impacts, sweeps, formant shots, reverse swells and zaps; the composer places them at the form's boundaries -- a riser over the last eight bars of a buildup, ending exactly on the drop, the formant shot on the last beat of the pre-drop break, a descending sweep into the drop.

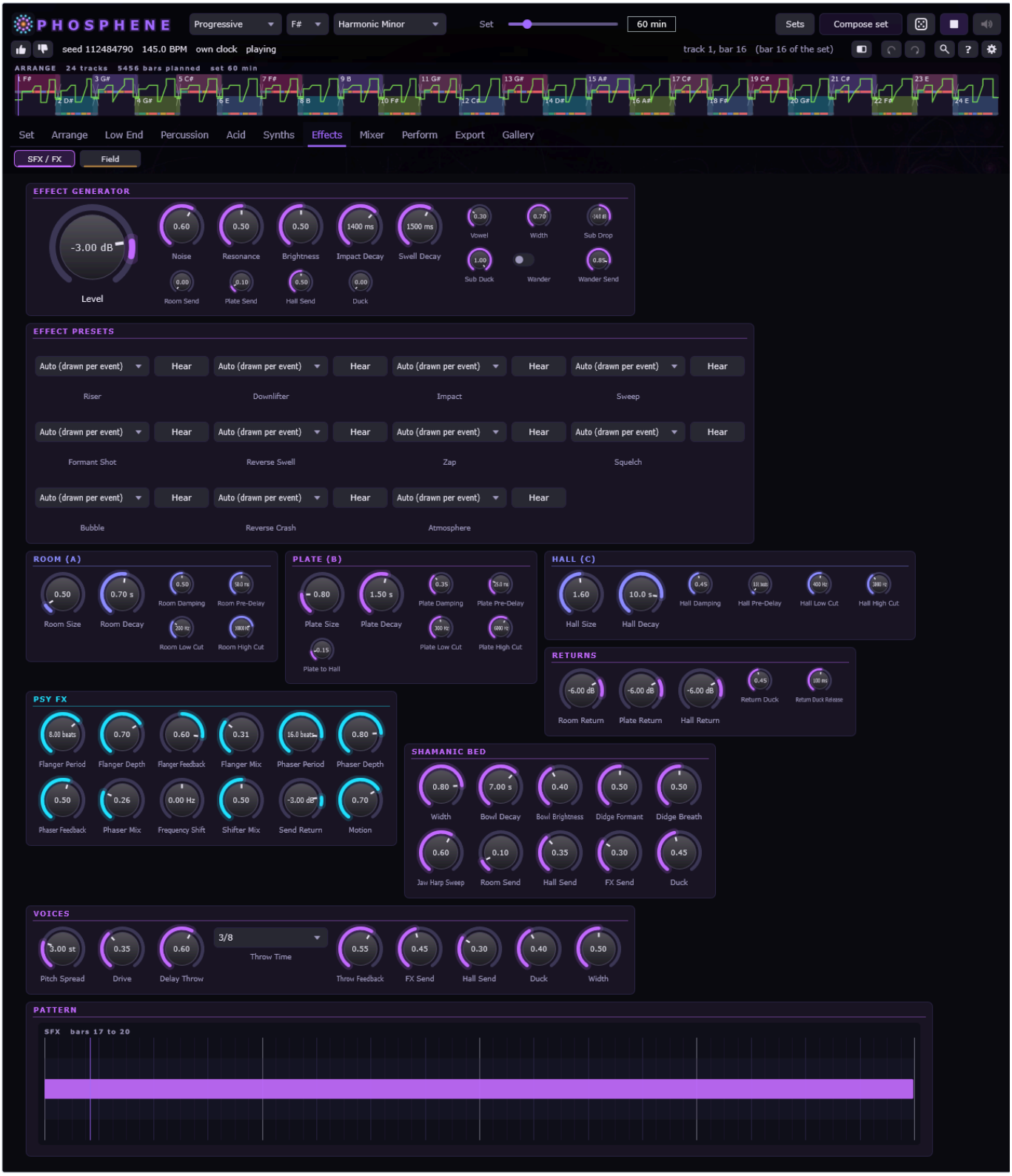
Short one-shots -- zaps, squelches, bubbles, alien chatter -- fall at the ends of phrases, about one in eight bars of a groove and one in four of the climax; a groove is never left eight bars without an effect, and what fills such a gap is a sweep or a swell, never a zap.

Effect Presets: every family of the effect bank (2048 presets in eleven families: riser, downlifter, impact, sweep, formant shot, reverse swell, zap, squelch, bubble, reverse crash, atmosphere) has a chooser. Auto lets the generator draw a preset of the family for every event, never the same twice in a track; a number makes every event of that family play that preset. "Hear" plays the chosen preset in the running set (with Auto, one of the family at random); while the transport stands nothing sounds.

Space Shot: a recording of NASA's, up to 25 seconds -- a rocket launch, Sputnik's beep, Saturn's radio emissions, a quake on Mars -- played whole, once, on this strip: after the cut of a breakdown, now and then in a long intro, most in Full-On and Hi-Tech, and never within two bars of a spoken phrase. It comes from the field recordings (Field tab); without them it stays silent.

The tonal layer of an effect -- an atmosphere's held tone above all -- is mapped into the mode, and never onto the b2 or the leading tone: a minor seventh the mode lacks becomes the fifth, not the major seventh.

The two send effects are a short room and a long hall, both feedback delay networks. Their returns are ducked with the kick like everything else, and high-passed: a reverb tail in the bass register is mud.



The SFX / FX tab.

EFFECT GENERATOR

Control	Key	Range	Default
Level	`sfx.level`	-36 .. 6 dB	-3
Noise	`sfx.noise`	0 .. 1	0.6
Resonance	`sfx.resonance`	0 .. 1	0.5
Brightness	`sfx.brightness`	0 .. 1	0.5
Impact Decay	`sfx.impact_decay`	200 .. 4000 ms	1400
Swell Decay	`sfx.swell_decay`	200 .. 6000 ms	1500
Vowel	`sfx.vowel`	0 .. 1	0.3
Width	`sfx.width`	0 .. 1	0.7

Sub Drop	`sfx.sub_level`	-36 .. 6 dB	-14
Sub Duck	`sfx.sub_duck`	0 .. 1	1
Wander	`sfx.wander`	off / on	off
Wander Send	`sfx.wander_send`	0 .. 1	0.85
Room Send	`sfx.room_send`	0 .. 1	0
Plate Send	`sfx.plate_send`	0 .. 1	0.1
Hall Send	`sfx.hall_send`	0 .. 1	0.5
Duck	`sfx.duck`	0 .. 1	0

EFFECT PRESETS

Control	Key	Range	Default
Riser	`sfx.preset_riser`	0 .. 256	0
Downlifter	`sfx.preset_downlifter`	0 .. 128	0
Impact	`sfx.preset_impact`	0 .. 128	0
Sweep	`sfx.preset_sweep`	0 .. 256	0
Formant Shot	`sfx.preset_formant_shot`	0 .. 96	0
Reverse Swell	`sfx.preset_reverse_swell`	0 .. 256	0
Zap	`sfx.preset_zap`	0 .. 96	0
Squelch	`sfx.preset_squelch`	0 .. 128	0
Bubble	`sfx.preset_bubble`	0 .. 64	0
Reverse Crash	`sfx.preset_reverse_crash`	0 .. 128	0
Atmosphere	`sfx.preset_atmosphere`	0 .. 512	0

ROOM (A)

Control	Key	Range	Default
Room Size	`fx.room_size`	0.3 .. 3	0.5
Room Decay	`fx.room_decay`	0.1 .. 4 s	0.7
Room Damping	`fx.room_damping`	0 .. 1	0.5
Room Pre-Delay	`fx.room_pre_delay`	0 .. 100 ms	50
Room Low Cut	`fx.room_low_cut`	100 .. 1000 Hz	200
Room High Cut	`fx.room_high_cut`	1000 .. 20000 Hz	10000

PLATE (B)

Control	Key	Range	Default
Plate Size	`fx.plate_size`	0.3 .. 3	0.8
Plate Decay	`fx.plate_decay`	0.2 .. 8 s	1.5
Plate Damping	`fx.plate_damping`	0 .. 1	0.35
Plate Pre-Delay	`fx.plate_pre_delay`	0 .. 100 ms	25
Plate Low Cut	`fx.plate_low_cut`	100 .. 1000 Hz	300
Plate High Cut	`fx.plate_high_cut`	1000 .. 20000 Hz	6000
Plate to Hall	`fx.plate_to_hall`	0 .. 1	0.15

HALL (C)

Control	Key	Range	Default
Hall Size	`fx.hall_size`	0.3 .. 3	1.6
Hall Decay	`fx.hall_decay`	0.3 .. 20 s	10
Hall Damping	`fx.hall_damping`	0 .. 1	0.45
Hall Pre-Delay	`fx.hall_pre_delay`	0 .. 0.5 beats	0.0125
Hall Low Cut	`fx.low_cut`	150 .. 1000 Hz	400
Hall High Cut	`fx.high_cut`	1000 .. 20000 Hz	3000

RETURNS

Control	Key	Range	Default
Room Return	`fx.room_return`	-36 .. 6 dB	-6
Plate Return	`fx.plate_return`	-36 .. 6 dB	-6
Hall Return	`fx.hall_return`	-36 .. 6 dB	-6
Return Duck	`fx.return_duck`	0 .. 1	0.45
Return Duck Release	`fx.return_duck_release`	10 .. 800 ms	100

PSY FX

Control	Key	Range	Default
Flanger Period	`psyfx.flanger_beats`	0.5 .. 32 beats	8
Flanger Depth	`psyfx.flanger_depth`	0 .. 1	0.7
Flanger Feedback	`psyfx.flanger_feedback`	-0.9 .. 0.9	0.6
Flanger Mix	`psyfx.flanger_mix`	0 .. 1	0.35
Phaser Period	`psyfx.phaser_beats`	0.5 .. 32 beats	16
Phaser Depth	`psyfx.phaser_depth`	0 .. 1	0.8
Phaser Feedback	`psyfx.phaser_feedback`	0 .. 0.9	0.5
Phaser Mix	`psyfx.phaser_mix`	0 .. 1	0.3
Frequency Shift	`psyfx.shift_hz`	-500 .. 500 Hz	0
Shifter Mix	`psyfx.shift_mix`	0 .. 1	0.5
Send Return	`psyfx.return`	-36 .. 6 dB	-3
Motion	`psyfx.motion`	0 .. 1	0.7

SHAMANIC BED

Control	Key	Range	Default
Width	`texture.width`	0 .. 1	0.8
Bowl Decay	`texture.bowl_decay`	1 .. 20 s	7
Bowl Brightness	`texture.bowl_bright`	0 .. 1	0.4
Didge Formant	`texture.didge_formant`	0 .. 1	0.5
Didge Breath	`texture.didge_breath`	0 .. 1	0.5
Jaw Harp Sweep	`texture.jaw_sweep`	0 .. 1	0.6
Room Send	`texture.room_send`	0 .. 1	0.1
Hall Send	`texture.hall_send`	0 .. 1	0.35
FX Send	`texture.fx_send`	0 .. 1	0.3
Duck	`texture.duck`	0 .. 1	0.45

VOICES

Control	Key	Range	Default
Pitch Spread	`vocal.pitch`	0 .. 12 st	3
Drive	`vocal.drive`	0 .. 1	0.35
Delay Throw	`vocal.throw_send`	0 .. 1	0.6
Throw Time	`vocal.throw_beats`	1/16 | 1/8 | 3/16 | 1/4 | 3/8 | 1/2	3/8
Throw Feedback	`vocal.throw_feedback`	0 .. 0.9	0.55
FX Send	`vocal.fx_send`	0 .. 1	0.45
Hall Send	`vocal.hall_send`	0 .. 1	0.3
Duck	`vocal.duck`	0 .. 1	0.4
Width	`vocal.width`	0 .. 1	0.5

16. Field

The Field track plays field recordings: a place under the music -- a rainforest, a night full of insects, rain on a roof, a river, a cave, a machine hall, the wind on Mars. It is a sampler, and its recordings are the original files, not converted: 120 recordings in 30 categories (four each, the steadiest and most different of their category; generated with Stable Audio 3 for Noctuary) and NASA's sounds from space. They are an optional part of the installation (2 GB). Without them everything plays; only this track and the NASA shots are silent, and the Library line says so.

Where it plays: under the intro from its first bar (so the place changes with the track, over the DJ's blend), over a breakdown, in the outro up to the blend, and -- in Dark Forest, rarely in Progressive -- sixteen quiet bars under a long groove. Never in a build-up or a drop. How often follows the style: Dark Forest always has its place, Hi-Tech least. compose.field_density scales it.

What plays: with A Category on Auto the composer draws a preset for every track by its style -- forest, night, insects and swamps for Dark Forest, sea, fire and ritual for Goa, wind, rain and water for Progressive, machines and radio for Hi-Tech, space for Full-On -- and sends it as it does the synths' presets: the knobs stay as they are. Choose a category and the knobs play, in every track.

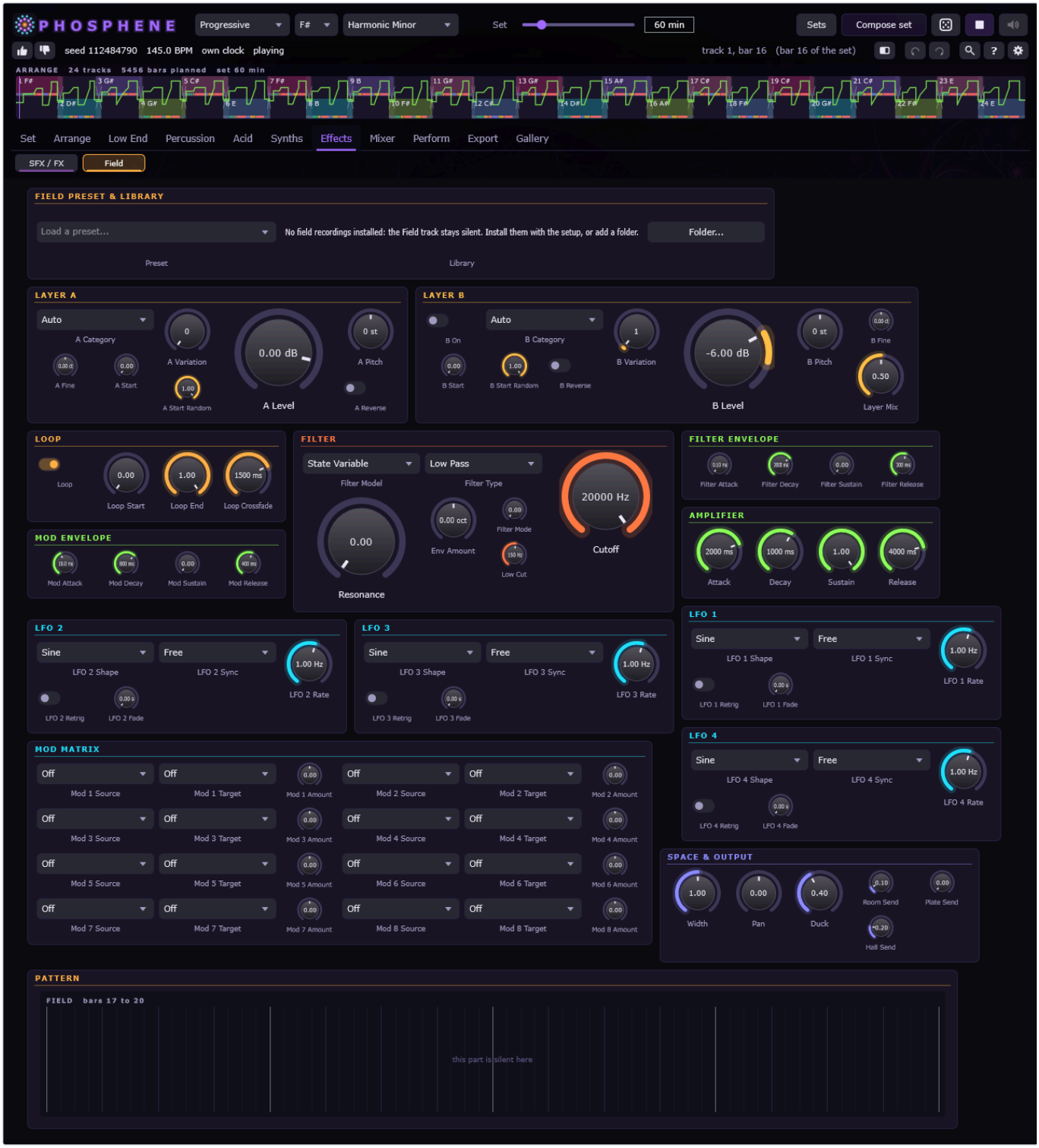
Preset: loads one of the factory presets into the knobs -- one per recording (by category) and the scenes, two recordings as one place (Jungle Night: a night and its insects; Swamp: the wetland and its mud). The category is then no longer Auto; set it back to hand the choice to the composer again. Library: how many recordings were found, and where. Folder...: a folder of your own recordings beside the installed ones -- FLAC files named fr-<category>-<anything>.flac (Noctuary's whole library of 2938 fits as it is); cancel the dialog to remove it.

Layer A and B: each a recording, chosen by category and variation (the variations of a category are its files in name order), with its level, its pitch (semitones and cents, by resampling), where it starts and how far that start may wander per event, and its direction. B is off by default; Layer Mix weighs the two (the middle plays both whole).

Loop: over any stretch of the recording (Loop Start to Loop End), joined by a crossfade of its own length -- over the last stretch before the loop's end a second read head runs from the loop's start, and the two are mixed with equal power, because two stretches of a field recording are uncorrelated and keep their loudness only so. What is heard at the seam is always a blend, never a cut, so a recording that was not made as a loop plays as one. Without the loop a layer ends with its recording.

Filter, Filter Envelope, Amplifier and the modulation block are the synths' (Filters and modulation); the matrix reaches pitch, cutoff, resonance, the filter's mode, level and pan. The Low Cut (24 dB per octave) keeps the recordings out of the range that belongs to kick and bass -- wind and water have most of their power there. The long amp envelope (2 s in, 4 s out) is how a place enters and leaves.

The recordings are loaded when a track is planned, a track ahead of their first note, and kept in memory; an offline render waits for them, so it is the same every time.



The Field tab.

LAYER A

Control	Key	Range
A Category	`field.a_category`	Auto | Rainforest | Night Country | Insects | Foliage | Wetland | Mud Bubbles | Cave | River | Sea | Rain on Land | Rain on Roof | Wind | Underwater | Ice and Snow |
A Variation	`field.a_variation`	0 .. 63
A Level	`field.a_level`	-36 .. 6 dB
A Pitch	`field.a_pitch`	-24 .. 24 st
A Fine	`field.a_fine`	-100 .. 100 ct
A Start	`field.a_start`	0 .. 1
A Start Random	`field.a_start_random`	0 .. 1

A Reverse	`field.a_reverse`	off / on

LAYER B

Control	Key	Range
B On	`field.b_on`	off / on
B Category	`field.b_category`	Auto | Rainforest | Night Country | Insects | Foliage | Wetland | Mud Bubbles | Cave | River | Sea | Rain on Land | Rain on Roof | Wind | Underwater | Ice and Snow |
B Variation	`field.b_variation`	0 .. 63
B Level	`field.b_level`	-36 .. 6 dB
B Pitch	`field.b_pitch`	-24 .. 24 st
B Fine	`field.b_fine`	-100 .. 100 ct
B Start	`field.b_start`	0 .. 1
B Start Random	`field.b_start_random`	0 .. 1
B Reverse	`field.b_reverse`	off / on
Layer Mix	`field.layer_mix`	0 .. 1

LOOP

Control	Key	Range	Default
Loop	`field.loop`	off / on	on
Loop Start	`field.loop_start`	0 .. 1	0
Loop End	`field.loop_end`	0 .. 1	1
Loop Crossfade	`field.loop_xfade`	10 .. 10000 ms	1500

FILTER

Control	Key	Range	Default
Filter Model	`field.filter_model`	State Variable | Moog Ladder | Prophet (SSM2040) | Juno (IR3109) | Oberheim SEM | Xpander | Diode Ladder | Korg35 (MS-20) | Polivoks | Wasp	State Variable
Filter Type	`field.filter_type`	Low Pass | Band Pass | High Pass | Notch	Low Pass
Cutoff	`field.cutoff`	40 .. 20000 Hz	20000
Resonance	`field.resonance`	0 .. 1	0
Env Amount	`field.env_amount`	-8 .. 8 oct	0
Filter Mode	`field.filter_mode`	0 .. 1	0
Low Cut	`field.low_cut`	20 .. 1000 Hz	150

FILTER ENVELOPE

Control	Key	Range	Default
Filter Attack	`field.filt_attack`	0.1 .. 4000 ms	0.1
Filter Decay	`field.filter_decay`	5 .. 30000 ms	2000
Filter Sustain	`field.filt_sustain`	0 .. 1	0
Filter Release	`field.filt_release`	5 .. 8000 ms	300

AMPLIFIER

Control	Key	Range	Default
Attack	`field.amp_attack`	1 .. 30000 ms	2000
Decay	`field.amp_decay`	5 .. 30000 ms	1000
Sustain	`field.amp_sustain`	0 .. 1	1
Release	`field.amp_release`	5 .. 30000 ms	4000

MOD ENVELOPE

Control	Key	Range	Default
Mod Attack	`field.menv_attack`	0.1 .. 8000 ms	10
Mod Decay	`field.menv_decay`	5 .. 12000 ms	800
Mod Sustain	`field.menv_sustain`	0 .. 1	0
Mod Release	`field.menv_release`	5 .. 12000 ms	400

LFO 1

Control	Key	Range	Default
LFO 1 Shape	`field.lfo1_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 1 Sync	`field.lfo1_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 1 Rate	`field.lfo1_rate`	0.01 .. 40 Hz	1
LFO 1 Retrig	`field.lfo1_retrig`	off / on	off
LFO 1 Fade	`field.lfo1_fade`	0 .. 8 s	0

LFO 2

Control	Key	Range	Default
LFO 2 Shape	`field.lfo2_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 2 Sync	`field.lfo2_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 2 Rate	`field.lfo2_rate`	0.01 .. 40 Hz	1
LFO 2 Retrig	`field.lfo2_retrig`	off / on	off
LFO 2 Fade	`field.lfo2_fade`	0 .. 8 s	0

LFO 3

Control	Key	Range	Default
LFO 3 Shape	`field.lfo3_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 3 Sync	`field.lfo3_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 3 Rate	`field.lfo3_rate`	0.01 .. 40 Hz	1
LFO 3 Retrig	`field.lfo3_retrig`	off / on	off
LFO 3 Fade	`field.lfo3_fade`	0 .. 8 s	0

LFO 4

Control	Key	Range	Default
LFO 4 Shape	`field.lfo4_shape`	Sine | Triangle | Saw Up | Saw Down | Square | Sample & Hold | Smooth Random	Sine
LFO 4 Sync	`field.lfo4_sync`	Free | 4 Bars | 2 Bars | 1 Bar | 1/2 | 1/4 | 1/8 | 1/16 | 1/4 T | 1/8 T	Free
LFO 4 Rate	`field.lfo4_rate`	0.01 .. 40 Hz	1
LFO 4 Retrig	`field.lfo4_retrig`	off / on	off
LFO 4 Fade	`field.lfo4_fade`	0 .. 8 s	0

MOD MATRIX

Control	Key	Range	Default
Mod 1 Source	`field.mx1_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 1 Target	`field.mx1_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 1 Amount	`field.mx1_amount`	-1 .. 1	0
Mod 2 Source	`field.mx2_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 2 Target	`field.mx2_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 2 Amount	`field.mx2_amount`	-1 .. 1	0
Mod 3 Source	`field.mx3_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 3 Target	`field.mx3_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 3 Amount	`field.mx3_amount`	-1 .. 1	0
Mod 4 Source	`field.mx4_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 4 Target	`field.mx4_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 4 Amount	`field.mx4_amount`	-1 .. 1	0
Mod 5 Source	`field.mx5_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 5 Target	`field.mx5_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 5 Amount	`field.mx5_amount`	-1 .. 1	0
Mod 6 Source	`field.mx6_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 6 Target	`field.mx6_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 6 Amount	`field.mx6_amount`	-1 .. 1	0
Mod 7 Source	`field.mx7_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 7 Target	`field.mx7_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off
Mod 7 Amount	`field.mx7_amount`	-1 .. 1	0
Mod 8 Source	`field.mx8_src`	Off | LFO 1 | LFO 2 | LFO 3 | LFO 4 | Mod Env | Filter Env | Velocity | Key | Random	Off
Mod 8 Target	`field.mx8_dst`	Off | Pulse Width | Cutoff | Resonance | Filter Mode | Level	Off

Mod 8 Amount	`field.mx8_amount`	-1 .. 1	0

SPACE & OUTPUT

Control	Key	Range	Default
Width	`field.width`	0 .. 2	1
Pan	`field.pan`	-1 .. 1	0
Duck	`field.duck`	0 .. 1	0.4
Room Send	`field.room_send`	0 .. 1	0.1
Plate Send	`field.plate_send`	0 .. 1	0
Hall Send	`field.hall_send`	0 .. 1	0.2

17. Mixer / Master

A channel strip per part: its pan (the six voices), its sends to the room, the hall and its delay or effects bus, its duck under the kick, the mute ("M") and the fader, beside a meter of what the part puts into the mix -- after fader, trance gate and duck, before the master. The green-to-red bar is the RMS level, the line above it the peak, held for a second and a half; the figure under the meter is that peak in dBFS. The knobs of a strip are the same parameters its synth page shows. Kick and bass have no strip level of their own: their fader is the synth's level.

The fader stays where you put it. Where the composer plays a part at another level -- the level match that holds every track's parts to one loudness, a preset's level correction -- a bright tick beside the fader shows the level that plays, joined to the fader's cap by a line.

Beside the strips the track gain and the sidechain's times, and the master chain: bus compressor, mono bass, a two-times oversampled knee clipper, a lookahead true-peak limiter and a safety clip, metered to ITU-R BS.1770.

Auto Gain renders four four-bar windows of the track through the whole master chain, measures them and offsets the master gain so that the output meets the loudness target. It runs when a track is planned, so it uses the knobs as they stand at that moment.

Depth comes in three planes, and three rooms. The front -- kick, bass and the lead -- is dry, the lead with a little of the near room (A: short, early reflections, a 50 ms pre-delay). The middle -- acid, counter, arp and stab -- sends to the plate (B, the main room: 1.5 s in a drop, 4 s in a breakdown, 25 ms pre-delay). The back -- pads and drone -- sends to the plate as well, more of it, and the plate in turn feeds the hall (C, the far room: ten seconds, 400 Hz .. 3 kHz), which the bed and the voices reach directly. The far room is heard only in intros and breakdowns: in a drop there are only A and B. Pre-delays are distance cues, not tempo values: near sources get the long one (their direct sound arrives well before the first reflection), far ones almost none. Each room has its own send filter (the Rooms group on the FX page).

Every voice carries a Distance (0 near .. 1 far, its own plane by default: the lead 0, the middle 0.5, pad and drone 1). Moving it moves all the cues of distance at once -- level, a low pass, the plate send and the width -- so a voice is never near by one cue and far by another. The composer uses it: the lead steps back in a breakdown and comes forward in the drop, and in the main breakdown the counter-lead arrives from far away over sixteen bars. Pads and drone also move slowly on their own (Slow Movement: cutoff, colour and width on periods of 13, 21 and 34 seconds, in golden-mean ratios, so no pattern is heard to come round).

Every strip gives way to the kick by what its plane needs: the bass 4 dB, the lines about 2 dB and back within 65 ms (Duck Release (Lines)), the back plane 5 to 6 dB within 125 ms (Duck Release), the returns 5 dB within 100 ms (Return Duck Release). The effects do not duck: an impact lands on the drop's kick. The lead drives two ducks of its own: the counter steps back while the lead plays (Counter under Lead), and so does the pad's presence band between 500 Hz and 3 kHz, for as long as a lead note sounds (Pad Band under Lead).

The composer mixes each phase differently. A breakdown stands 7 dB under the drop before it (after the master's compressor and limiter that is about 5 dB of loudness), with the pad at full width and a little louder, the lead further back and the far room open; a buildup narrows the pad, pulls the plate down to 1.2 seconds, fades the far room out and lifts the lines' high pass, and the drop opens it all again within a bar -- the lead at the front, the pad at its full width again, the pad and drone ducking deeper. In the intro and the breakdown, where pad and drone are often the only tonal voices, their unison stays within about six cents: held, a wider stack beats. While the bass plays, the pad leaves the third to the lead. On the last beat before every eight-bar change of a groove or a drop the effects are thrown into the plate for that one beat.

Monitor (Master page): Mono, Sub (a low pass at 80 Hz, where the kick-bass gap and the tuning are judged) or Side, after the meter -- a listening switch, it changes nothing the loudness readings see.



The Mixer / Master tab.

CHANNELS

Control	Key	Range	Default
Level	`kick.level`	-36 .. 6 dB	-6
Kick Mute	`mix.kick_mute`	off / on	off
Level	`bass.level`	-36 .. 6 dB	-3
Bass Mute	`mix.bass_mute`	off / on	off
Perc Level	`mix.perc_level`	-24 .. 12 dB	1.2
Perc Mute	`mix.perc_mute`	off / on	off
Perc Room	`mix.perc_room`	0 .. 1	0.12
Perc Hall	`mix.perc_hall`	0 .. 1	0
Acid Level	`mix.acid_level`	-24 .. 12 dB	0
Acid Mute	`mix.acid_mute`	off / on	off
Room Send	`acid.room_send`	0 .. 1	0.1
Hall Send	`acid.hall_send`	0 .. 1	0
Delay Send	`acid.delay_send`	0 .. 1	0.25
Duck	`acid.duck`	0 .. 1	0.15
Lead Level	`mix.lead_level`	-24 .. 12 dB	-6
Lead Mute	`mix.lead_mute`	off / on	off
Pan	`lead.pan`	-1 .. 1	-0.2
Room Send	`lead.room_send`	0 .. 1	0.1
Hall Send	`lead.hall_send`	0 .. 1	0
Delay Send	`lead.delay_send`	0 .. 1	0.3
Duck	`lead.duck`	0 .. 1	0.2
Counter Level	`mix.counter_level`	-24 .. 12 dB	5
Counter Mute	`mix.counter_mute`	off / on	off

Pan	`counter.pan`	-1 .. 1	0.2
Room Send	`counter.room_send`	0 .. 1	0
Hall Send	`counter.hall_send`	0 .. 1	0
Delay Send	`counter.delay_send`	0 .. 1	0.4
Duck	`counter.duck`	0 .. 1	0.2
Arp Level	`mix.arp_level`	-24 .. 12 dB	-5
Arp Mute	`mix.arp_mute`	off / on	off
Pan	`arp.pan`	-1 .. 1	0
Room Send	`arp.room_send`	0 .. 1	0
Hall Send	`arp.hall_send`	0 .. 1	0
Delay Send	`arp.delay_send`	0 .. 1	0.35
Duck	`arp.duck`	0 .. 1	0.2
Stab Level	`mix.stab_level`	-24 .. 12 dB	-3
Stab Mute	`mix.stab_mute`	off / on	off
Pan	`stab.pan`	-1 .. 1	0
Room Send	`stab.room_send`	0 .. 1	0
Hall Send	`stab.hall_send`	0 .. 1	0
Delay Send	`stab.delay_send`	0 .. 1	0.45
Duck	`stab.duck`	0 .. 1	0.2
Pad Level	`mix.pad_level`	-24 .. 12 dB	0
Pad Mute	`mix.pad_mute`	off / on	off
Pan	`pad.pan`	-1 .. 1	0
Room Send	`pad.room_send`	0 .. 1	0
Hall Send	`pad.hall_send`	0 .. 1	0
Delay Send	`pad.delay_send`	0 .. 1	0
Duck	`pad.duck`	0 .. 1	0.5
Drone Level	`mix.drone_level`	-24 .. 12 dB	0
Drone Mute	`mix.drone_mute`	off / on	off
Pan	`drone.pan`	-1 .. 1	0
Room Send	`drone.room_send`	0 .. 1	0
Hall Send	`drone.hall_send`	0 .. 1	0
Delay Send	`drone.delay_send`	0 .. 1	0
Duck	`drone.duck`	0 .. 1	0.45
SFX Level	`mix.sfx_level`	-24 .. 12 dB	4.5
SFX Mute	`mix.sfx_mute`	off / on	off
Room Send	`sfx.room_send`	0 .. 1	0
Hall Send	`sfx.hall_send`	0 .. 1	0.5
Duck	`sfx.duck`	0 .. 1	0
Texture Level	`mix.texture_level`	-24 .. 12 dB	9
Texture Mute	`mix.texture_mute`	off / on	off
Room Send	`texture.room_send`	0 .. 1	0.1
Hall Send	`texture.hall_send`	0 .. 1	0.35
FX Send	`texture.fx_send`	0 .. 1	0.3
Duck	`texture.duck`	0 .. 1	0.45
Vocal Level	`mix.vocal_level`	-24 .. 12 dB	7
Vocal Mute	`mix.vocal_mute`	off / on	off
Hall Send	`vocal.hall_send`	0 .. 1	0.3
FX Send	`vocal.fx_send`	0 .. 1	0.45
Delay Throw	`vocal.throw_send`	0 .. 1	0.6
Duck	`vocal.duck`	0 .. 1	0.4
Field Level	`mix.field_level`	-24 .. 12 dB	-14
Field Mute	`mix.field_mute`	off / on	off
Pan	`field.pan`	-1 .. 1	0
Room Send	`field.room_send`	0 .. 1	0.1
Hall Send	`field.hall_send`	0 .. 1	0.2
Duck	`field.duck`	0 .. 1	0.4

TRACK & SIDECHAIN

Control	Key	Range	Default
Track Gain	`mix.track_gain`	-12 .. 12 dB	0
Duck Attack	`mix.duck_attack`	0.5 .. 30 ms	2
Duck Hold	`mix.duck_hold`	0 .. 200 ms	20
Duck Release	`mix.duck_release`	10 .. 800 ms	125
Duck Release (Lines)	`mix.duck_release_lines`	10 .. 800 ms	65
Counter under Lead	`mix.counter_duck`	0 .. 1	0.2
Pad Band under Lead	`mix.pad_lead_duck`	0 .. 12 dB	3

GAIN

Control	Key	Range	Default
Gain	`master.gain`	-24 .. 24 dB	0
Ceiling	`master.ceiling`	-12 .. 0 dBTP	-3
Clip	`master.clip`	off / on	on

COMPRESSOR

Control	Key	Range	Default
Comp Threshold	`master.comp_threshold`	-40 .. 0 dB	-8
Comp Ratio	`master.comp_ratio`	1 .. 10 :1	1.5
Comp Knee	`master.comp_knee`	0 .. 24 dB	6
Comp Attack	`master.comp_attack`	0.1 .. 100 ms	30
Comp Release	`master.comp_release`	10 .. 2000 ms	200

OUTPUT

Control	Key	Range	Default
Mono Bass	`master.mono_bass`	40 .. 300 Hz	120
Limiter	`master.limiter`	off / on	on
Limiter Release	`master.limiter_release`	2 .. 1000 ms	20
Target Loudness	`master.target_lufs`	-20 .. -4 LUFS	-9
Auto Gain	`master.auto_gain`	off / on	on
Clipper	`master.clipper`	off / on	on
Clipper Threshold	`master.clipper_threshold`	-6 .. 6 dB	0

MONITOR

Control	Key	Range	Default
Monitor	`master.monitor`	Normal | Mono | Sub | Side	Normal

TONE

Control	Key	Range	Default
Resonance	`master.resonance`	0 .. 1	0.4
Auto Tone	`master.auto_tone`	off / on	on
Tone Low Mid	`master.tone_low_mid`	-6 .. 6 dB	0
Tone Mid	`master.tone_mid`	-6 .. 6 dB	0
Tone Presence	`master.tone_presence`	-6 .. 6 dB	0
Tone Air	`master.tone_air`	-6 .. 6 dB	0

REFERENCE

Control	Key	Range	Default
Image	`master.image`	0 .. 1	1
Balance	`master.balance`	0 .. 1	1

18. Perform

Four macros, meant for a hand rather than a mouse.

They are immediate parameter moves rather than score events, and that is measured rather than convenient: the engine's control ring is first in, first out and the composer keeps it filled eight bars ahead, so an event pushed into it now would not be heard for about thirteen seconds. A macro therefore writes the knob itself, as an offset in the knob's normalised domain, and puts it back exactly where it was when you let go. The composer's own ramps keep riding on top.

Drop-out is a press, not a hold: it takes the kick and the bass out and lets them back in on the next bar line. Stutter is held. Filter Sweep and Gate Depth stay where you leave them, and a double click puts them back to neutral.

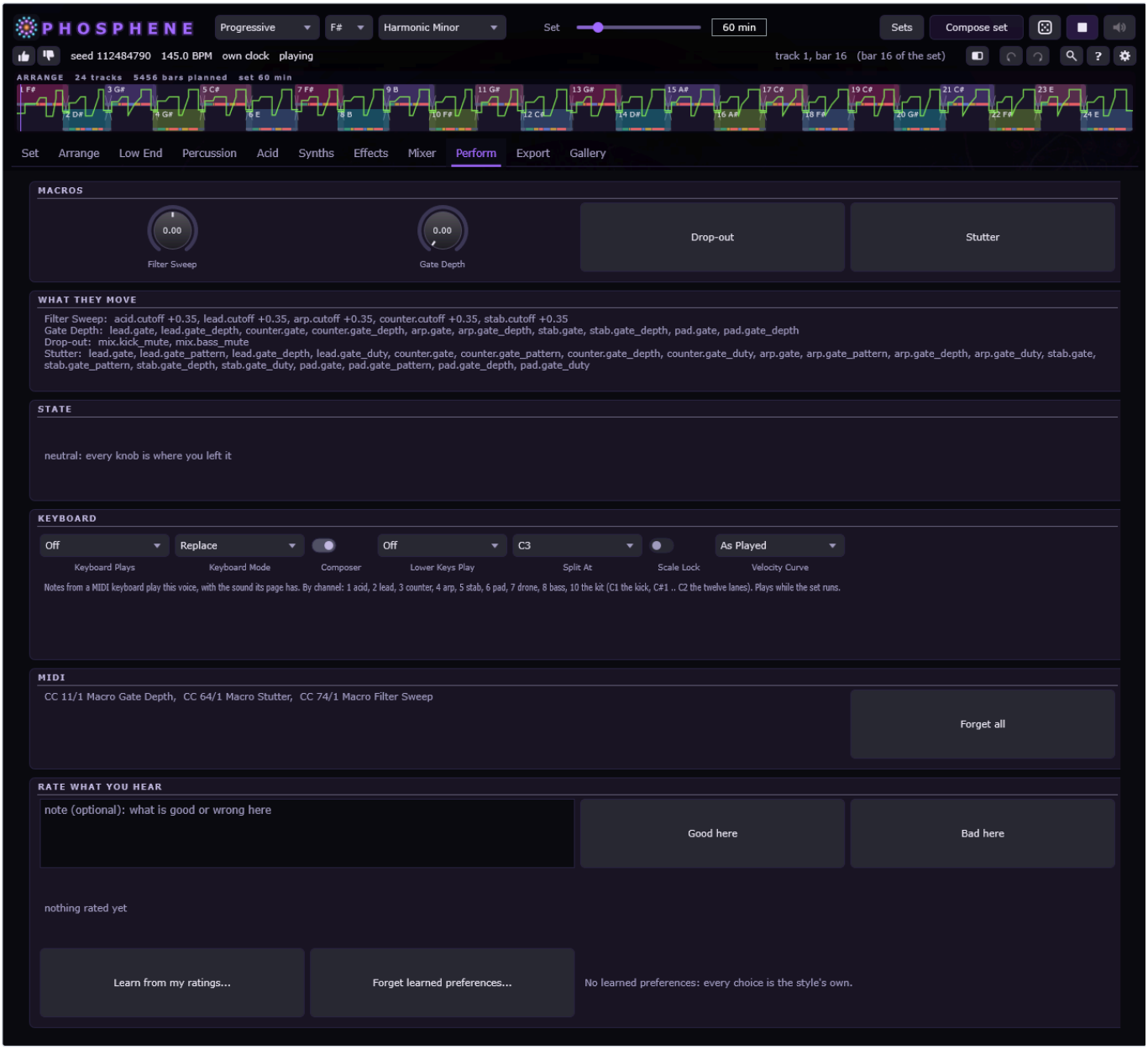
Keyboard: notes from a MIDI keyboard can play one voice -- the acid, any of the six melodic voices, the bass or the kit (C1 the kick, C#1 to C2 the twelve lanes), or "By channel" (channel 1 the acid, 2 the lead ... 7 the drone, 8 the bass, 10 the kit) -- with the sound that voice's tab has. Replace leaves the generator's notes of that voice out, Layer plays over them. Composer off leaves every generated note out: then only what you play sounds, through the mix and the effects as composed. The keyboard plays while the set runs; an export always plays what was composed.

Three options of the Keyboard group, each off until you choose it: Lower Keys Play splits the keyboard -- the keys below Split At (C1 to C5, C3 to begin with) play the voice chosen there, the keys from it upwards the voice of Keyboard Plays; Scale Lock moves a key outside the track's key and mode to the nearest note inside it, so no wrong note can sound; Velocity Curve shapes the touch -- As Played, Soft (a light touch is enough), Hard (it takes a firm one) or Fixed (every note at 100, for pads that send no velocity). A key's release always ends the note its press started, even when an option changed while it was held.

MIDI: every knob, switch and chooser, and the four macros, can be put on a hardware controller. Right click the control, choose MIDI Learn and turn the knob on your controller. The MIDI group lists every binding and forgets them all at once. Without learning, controller 74 is the filter sweep, 11 the gate depth and 64 the stutter -- the controllers every generator of the family listens to.

Headset: while a Meta Quest sends its hands (the Quest app's bridge mode), the Perform tab shows what they do -- the left pinch play and stop, the right pinch the drop-out, held the stutter, both hands the next track, the left hand the filter sweep, the right the gate depth (The panel).

Rate what you hear: Good here and Bad here write a verdict with a note into your ratings. Learn from my ratings fits your taste to them and, once you confirm, weights the choices the style leaves open towards what you liked. The genre's rules always stay above your taste.



The Perform tab.

Filter Sweep

Opens or closes the filters of acid, lead, counter-lead, arp and stab together, by up to a third of their range in each direction. Centre is neutral.

Moves: `acid.cutoff +0.35, lead.cutoff +0.35, arp.cutoff +0.35, counter.cutoff +0.35, stab.cutoff +0.35`

Gate Depth

Switches the trance gate on for lead, counter-lead, arp, stab and pad and sets its depth; at 1.0 the gate closes completely between its steps.

Moves: `lead.gate = 1, lead.gate_depth = 1, counter.gate = 1, counter.gate_depth = 1, arp.gate = 1, arp.gate_depth = 1, stab.gate = 1, stab.gate_depth = 1, pad.gate = 1, pad.gate_depth = 1`

Drop-out (press)

Takes kick and bass out until the next bar line and lets them back in on the downbeat. One press, one bar.

Moves: `mix.kick_mute = 1, mix.bass_mute = 1`

Stutter (press)

Held: lead, counter-lead, arp, stab and pad run through the gate on sixteenths at full depth and the shortest duty. Phosphene has no buffer repeat, so this is the gate's stutter, not a tape one.

Moves: `lead.gate = 1, lead.gate_pattern = 0, lead.gate_depth = 1, lead.gate_duty = 0, counter.gate = 1, counter.gate_pattern = 0, counter.gate_depth = 1, counter.gate_duty = 0, arp.gate = 1, arp.gate_pattern = 0, arp.gate_depth = 1, arp.gate_duty = 0, stab.gate = 1, stab.gate_pattern = 0, stab.gate_depth = 1, stab.gate_duty = 0, pad.gate = 1, pad.gate_pattern = 0, pad.gate_depth = 1, pad.gate_duty = 0`

KEYBOARD

Control	Key	Range	Default
Keyboard Plays	`mix.keyboard_part`	Off | Acid | Lead | Counter | Arp | Stab | Pad | Drone | By channel | Bass | Kit	Off
Keyboard Mode	`mix.keyboard_mode`	Replace | Layer	Replace
Composer	`mix.composer`	off / on	on
Lower Keys Play	`mix.keyboard_lower`	Off | Acid | Lead | Counter | Arp | Stab | Pad | Drone | Bass | Kit	Off
Split At	`mix.keyboard_split`	C1 | C2 | C3 | C4 | C5	C3
Scale Lock	`mix.keyboard_scale`	off / on	off
Velocity Curve	`mix.keyboard_velocity`	As Played | Soft | Hard | Fixed	As Played

19. Export

What leaves the instrument, in one place, as every generator of the family has it (01.10.2026): Record (the red disc) writes the output as it plays into a WAV file until it is pressed again; Export MIDI writes the score of so many bars from the start, a channel per part; the disk writes the set -- the seed, the locks, the rerolls and every knob that differs from the factory -- as a .phosset, and the folder reads one and plays it. The score cues (OSC) send the set's beats, bars, sections and keys to a light show or a VJ program at the moment they are heard.

The keys: Ctrl+E exports the MIDI, Ctrl+S the set, Ctrl+O loads one.

Four things connect Phosphene to the software around it and to the rest of the family (02.10.2026). The outputs, Link and the jam are off until they are wanted, the MIDI goes nowhere until a track takes it, and only the jam -- as a follower -- changes what the music sounds like.

MIDI out: in a DAW the plugin also sends what it plays -- every note of every part at the moment it sounds, a channel per part as in the MIDI file of an export -- and leaves out a voice the keyboard plays instead (Replace). Record it on a MIDI track to keep a part as notes, or route it to another instrument to double it. A stop or a jump of the transport sends all notes off.

Outputs of their own: besides the main output the plugin has a stereo output for each of the stems -- the kick, the bass, the percussion, the acid, the lead, the counter, the arp, the stab, the pad, the drone, the effects, the texture, the voice, the field recordings and the returns. They are off to begin with; switched on in the DAW (in most hosts: the plugin's output configuration, or its multi-output version), each carries its part before the master, while the main output plays on as before.

Ableton Link (the standalone; Settings > Ableton Link): with another Link app on the network -- Live, Bitwig, a phone app, another generator of the family -- Phosphene takes the session's tempo, lines its bars up with the session's and starts and stops with it; alone in a session it offers its own tempo to whoever joins. In a DAW the host's transport leads, as before.

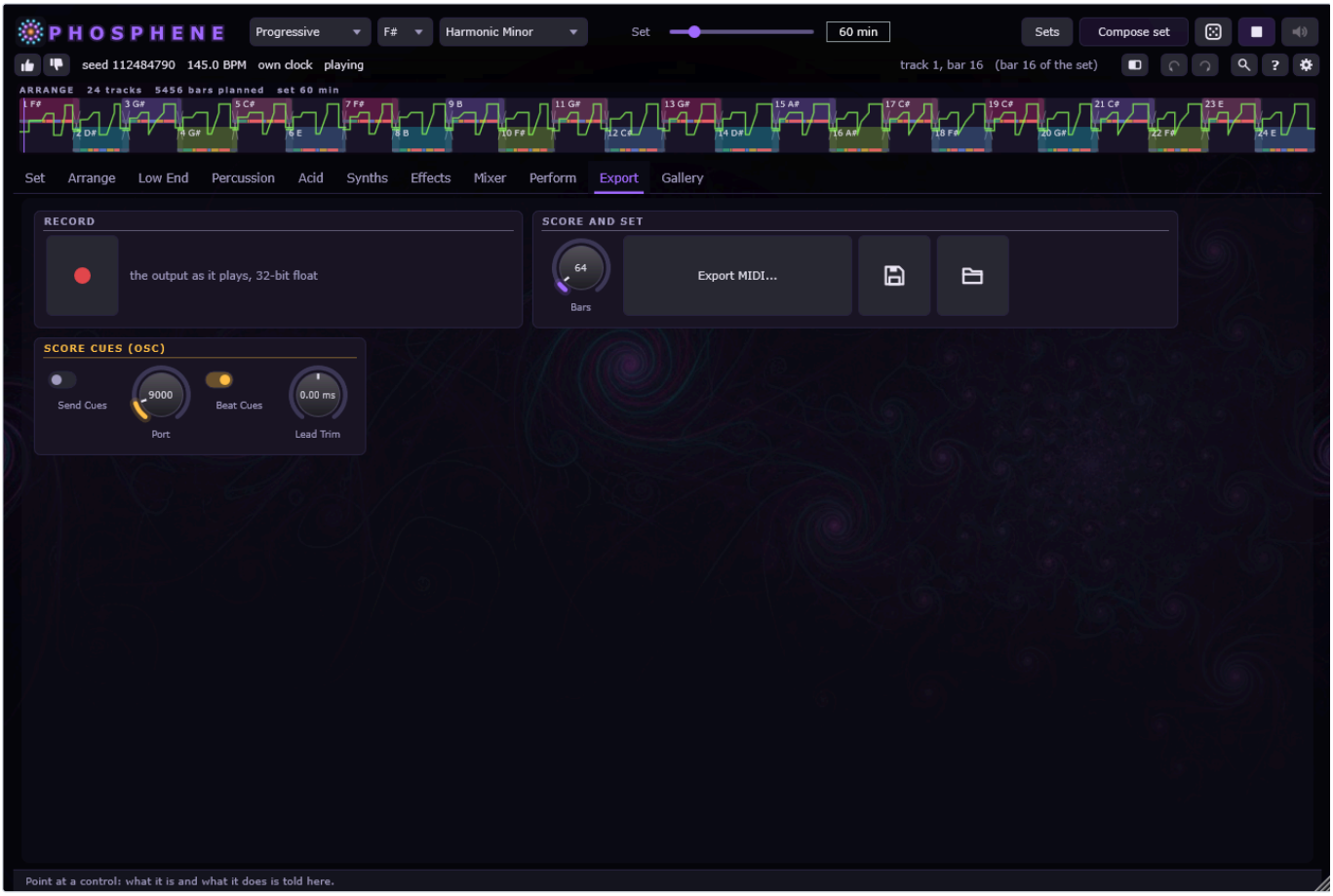
The family jam (Settings > Family jam, in the plugin and in the standalone; off to begin with): the five instruments of the family -- Totality, Parhelion, Ephemeris, Phosphene and Noctuary -- play as one band on the local network. One leads: its key, the energy of its sections, its breaks and its drops go out to the others. Those set to Follow take them up -- in Phosphene: the bass, the 303, the lead, the counter, the arp, the stab, the pad and the drone move to the leader's key while the kit and the effects play as written, and the set's walk holds its key; the Filter Sweep macro closes with a quiet section of the leader's; and in the leader's breaks the kick and the bass are out until its drop. A major mode of the leader's it leaves be: psytrance's palette has none. A new root comes in at the follower's next bar line, by the shortest way, the mode with its next track. Tempo and bars come from Ableton Link in the standalone (switch it on in each of them) or from the DAW's transport, so that a section the leader announces lands on the same bar everywhere. As the leader Phosphene says its set's key and each section of the set as it passes, with its energy. The jam stays on this network (UDP multicast to 239.255.77.77, port 9177, not past the router), instruments on one machine find each other as well, and a follower that has not heard its leader for four seconds goes back to its own key and its own form. The last line of the submenu says what it does: off, leading, or whom it follows.

On a Mac: every release has a build for Apple Silicon (macOS 12 or newer), the standalone, the VST3 and the Audio Unit (Logic, GarageBand, MainStage; passed by Apple's auval), built and tested on GitHub's machines. It is signed ad hoc, not notarized: macOS asks once (README-macOS.txt in the zip says how), and it has not yet been played on a real Mac -- reports are welcome.

On Linux: every release has an archive for x86-64 (glibc 2.35 or newer: Ubuntu 22.04, Debian 12, Fedora 36 and later) with the standalone, the VST3, the LV2 (Ardour, Carla, Reaper, Qtractor) and the renderer, built and tested on GitHub's Ubuntu machines and tried under WSL. README-Linux.txt in it says where everything goes: the VST3 into ~/.vst3, the LV2 into ~/.lv2, the standalone started where it lies, sound through ALSA or JACK (PipeWire serves both).

Demos: the release "demos" holds a track per style as an MP3 and one of them as a video, with pictures by KaleidoscopeEnhanced cut to the track's own score cues; Tools/demo/make_demos.py makes them all again (phos_render).

KaleidoscopeEnhanced understands the score cues of all five instruments as they come (since 02.10.2026) -- Totality's blocks and keys, Parhelion's sections, Ephemeris' phases, Phosphene's sections and drops, Noctuary's bars, keys and scenes -- and cuts its pictures to them, on a drop at once.



The Export tab.

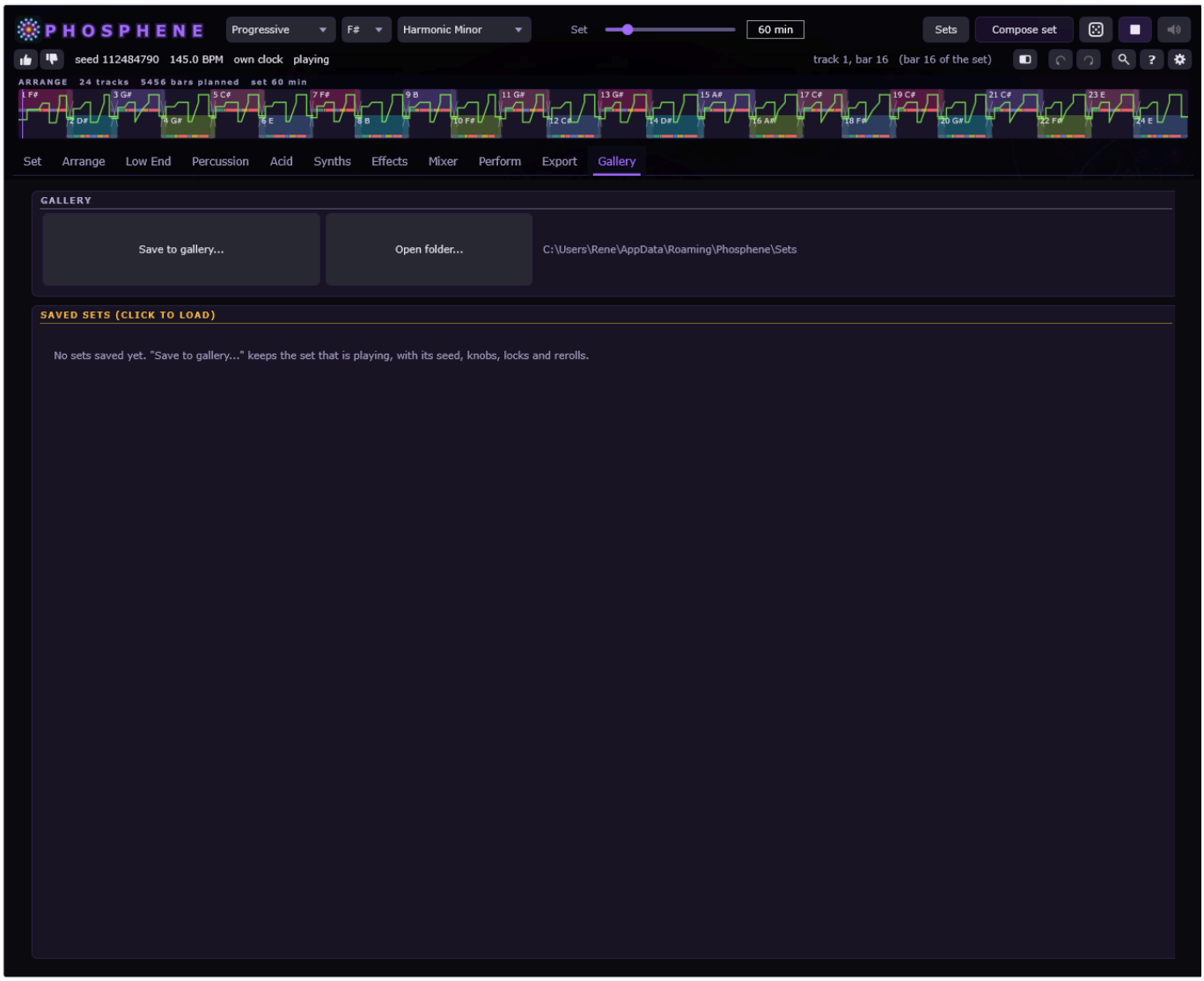
SCORE CUES (OSC)

Control	Key	Range	Default
Send Cues	`cue.send`	off / on	off
Port	`cue.port`	1024 .. 65535	9000
Beat Cues	`cue.beats`	off / on	on
Lead Trim	`cue.lead_ms`	-50 .. 50 ms	0

20. Gallery

Your saved sets, each with a picture of its form. Save to gallery keeps the set as it stands -- seed, style, arc, knobs, locks and rerolls -- and a click on an entry loads it back. The files are ordinary .phosset files in your Sets folder; Open folder shows it.

Phosphene Sets, a small program beside the instrument (in the Start menu, or PhospheneSets.exe in its folder), renders without the instrument: a set from each start style -- ten minutes to three hours, its arc shaped over that length -- and every set you saved (the gallery's and those in Documents/Phosphene), each with the length of its rendering; or a set's first track alone, at its own tempo, as the DJ export writes it. Choose one or several (Ctrl, Shift), the format (WAV or FLAC at 24 bits, Ogg Vorbis or MP3 at 320 kbit/s) and the folder (Documents/Phosphene/Renders at first), and Render: one after another, faster than real time, with the progress, the speed and the time left. A set never ends by itself, so its last twenty seconds are faded out. New dice composes everything anew; a saved set plays as it was saved.



The Gallery tab.

21. Filters and modulation

The melodic voices and the bass choose their filter's circuit with Filter Model. The first entry is the synth's own filter -- the voices' state-variable filter, the bass's ladder -- and the others are circuit models solved sample by sample: the Moog transistor ladder, the OTA cascades of the Prophet and the Juno, the Oberheim SEM, the Xpander's pole mixes, the diode ladder of the TB-303, the Korg35 of the MS-20, the Polivoks and the Wasp. Filter Mode morphs the SEM from low pass through notch to high pass, steps through the Xpander's eight responses and turns the Polivoks into a band pass. Switching models keeps the level: each is calibrated against the synth's own filter. The acid keeps its diode ladder -- that is the acid.

The filter envelope is a full ADSR: Filter Attack, Decay, Sustain and Release. With the attack at its minimum and the sustain at 0 it is the plain decay every older set used.

A third envelope (Mod Envelope), four LFOs and a modulation matrix of eight slots move the rest. An LFO runs free at its rate or synced to the beat (4 bars down to a sixteenth, and triplets), in seven shapes from sine to smooth random; Retrig restarts it with every note, Fade brings it in over a few seconds. A slot of the matrix reads a source -- an LFO, the mod or filter envelope, the velocity, the key or a random value drawn per note -- and moves a target by its amount: pitch, the second oscillator's pitch, pulse width, the wavetable position (the classic slow scan through a table), the FM index, cutoff, resonance, the filter mode, the level or the pan. The bass offers only what cannot break its lock to the kick (no pitch, no pan), the acid what suits a 303 line.

22. Sound presets

Every synth tab starts with a Sound group: the synth's presets in a chooser, grouped by kind, a disk for presets of your own, and the Own Sound switch.

Every track of a set plays a preset on every synth that is not on Own Sound: the composer chooses one by the track's style -- a Hoover or a Supersaw Anthem for Full-On, a Growl or a Screaming Glide for Dark Forest, a Glass Lead or a Psy Pluck for Progressive -- and never the same group as the two tracks before. While it plays, the tab shows it: the chooser reads "Group: Name (composer)" and the knobs stand at the preset's values. The sections still move some of them on top, the way a player rides a filter; those moves are not written into the knobs. The composer also evens out the presets' loudness: each carries a correction, measured against the synth's own default sound, that rides on the synth's Level on top of your setting, and the level match then holds every part of every track -- the first one included -- to the loudness the default sound would have there. Sound Presets on the Set page (or Sound Variation at 0) turns the composer's presets off, and every synth plays its knobs again, with the composer's recipes.

Own Sound keeps the composer out: the synth then sounds exactly as its tab says, in every track. Turning a knob of a synth that plays the composer's preset switches its Own Sound on, with the knobs where the preset put them, so the sound you start from is the one you heard. Choosing a preset from the chooser switches Own Sound on as well. A preset sets the synth's sound and leaves its level, its ducking, its sends to the rooms, its place in the image and -- on the melodic voices -- its high pass and trance gate alone, since those are where the mix and the form put the voice. Undo takes a preset back.

The live ring: where a knob plays away from where it stands -- a section riding the cutoff, drop 2's lift, a preset's level correction -- a thin bright arc on the knob's ring runs from the knob to the value that plays, with a dot where it plays. The modulation shows there too: an LFO or the mod envelope on the cutoff, the table position, the resonance, the filter mode, the pulse width, the FM index, the level or the pan moves the dot as the latest note's voice moves. A knob without it plays exactly what it shows.

The factory bank holds 1024 presets for each synth: sixteen groups -- Hoover, Acid Lead, Vocal Formant, Rolling, Squelchy, Classic 303, Full-On ... -- of sixty-four each. A group is a grid of eight adjectives from dark to bright (the lead's run from Umbral to Supernova) and eight nouns of its own (the Hoover's from Hoover to Titan), so "Umbral Mentasm" is the darkest of the Hoovers' second column and "Supernova Titan" the brightest of its last. Each preset brings the filter model its group suits and modulation from its group's recipes: a slow LFO on the cutoff, the wavetable scanned by an LFO or the mod envelope, a hoover's pitch dive, a vibrato that fades in. Your own presets are listed under User and live in the Presets folder of your Phosphene user folder.

23. The panel

Phosphene shares its panel with the other generators of the family -- Ephemeris, Totality and Parhelion (and, in part, Noctuary) -- so that what is learned on one holds on the others: the same places, the same keys, the same controllers, the same hands. Each keeps its own colours, its own letters and the picture behind its panel (Settings > Picture behind the panel takes it away).

The header's first row: the logo, the style, the key and the scale, what is composed and its length, Compose, the die (a new seed), Play and the speaker (mute). Its second row: Good here and Bad here -- a verdict on what plays, for your taste --, the status line, and at the right Undo, Redo, Help and the settings. Under the header the whole set at a glance; a click jumps there. The tabs come in one order everywhere: Set, Arrange, the instrument's own pages -- a page of several has small tabs of its own --, Mixer, Perform, Export, Gallery.

The unambiguous actions are icons since 03.10.2026, their words in the tooltips: the die draws a new seed, the triangle plays and the square stops, the speaker mutes (crossed out while muted), the disk saves and the folder loads, the red disc records (Export).

The ways around the panel (15.10.2026, the family's). The magnifier at the right of the header (Ctrl+K) opens the search: a few letters find any knob -- its page opens, its section or lane comes to the front and it lights up; a number after its name sets it ("cutoff 800") --, any preset of any synth, a page or an action. The line along the bottom of the window says what the control under the mouse is: its value now, its range and default, where the score plays it, what moves it, its MIDI controller and its key. Each synth's Sound group has a magnifier, the browser: by name, by group, the starred and the recently played; hold a row's triangle to hear it, double-click to take it. The name under the chooser carries a star while the knobs have moved from the preset chosen. A knob a voice's modulation moves wears an arc round it for every row of the matrix on it -- the span the row can move it over, in its source's colour; a right click lists those rows, each with its depth to drag and a button to take it out, and offers every source (Modulate with: a row in the first free slot, a quarter of the target's span). Held, the split square in the header (A|B) plays the presets as they were chosen; let go, the sound as it has been turned since. The first steps show the panel on the first start (Settings > First steps, again). The standalone's window comes back where it was left -- its place, its size, maximised or not --, and its audio settings offer ASIO (Windows).

A page taller than the window shows its groups in sections, one at a time, with a switch at its top right: a synth's sound and its modulation apart (Sound, Modulation), and where even those do not fit, cut further, each named after its first group (Filter, LFO, Matrix ...). At the window's usual size, 1280 by 860, no page scrolls. The overview above the tabs can be folded away in the settings; the pages then get its height.

A colour says what a control does, the same on every page and in every generator of the family: the sources, the filters, the envelopes, what moves by itself, the room and the mix. A knob the composer moves has a thin ring round it that shows where it stands at this moment. A double click sets a control to its default; a right click offers MIDI learn, forgetting the controller, and the default.

Undo and Redo take back and bring back what was changed -- a knob, a new seed, a reroll, a preset, a loaded set -- one step at a time; the button's tooltip names the step. Help opens this manual by topic, with the page in front, the keys and the headset as topics of their own. The settings hold the update check (it asks GitHub once a day whether a newer Phosphene is out; nothing else is sent), the picture behind the panel, the overview, the headset, the window's size, full screen, the keys and About.

```
Space                 play / stop
Ctrl+Z                undo
Ctrl+Y, Ctrl+Shift+Z  redo
F1                    help
F11                   full screen (the standalone); Esc leaves it and closes the help
Ctrl+S, Ctrl+O        save the set, open one
Ctrl+E                export
Ctrl+K                search: any control, preset, page or action
```

The controllers every generator of the family listens to without learning: controller 74 (brightness) is the filter sweep, 11 (the expression pedal) the gate depth, 64 (the sustain pedal) the stutter while it is held; the mod wheel is left free. A set with bindings of its own keeps them.

The headset. The controls of a Meta Quest's hands are shown only while a headset sends them -- or always, if the settings say so (Headset > Always show them); Off stops listening. The Quest app sends its hands to this computer in bridge mode: bridge_host = this computer's address in its phos.cfg (audio=0 leaves the headset silent), and Phosphene listens on port 9101. The hands mean the same in every generator of the family, on the headset alone and here:

```
left pinch            play / stop
both hands pinched    the next track
right pinch           drop-out: kick and bass gone until the next bar line
right pinch held      stutter, while it is held
left hand up, down    filter sweep: closed below the middle, open above
right hand up         gate depth
```

A pinch acts when it opens again, so a pinch of both hands is never also one of each; a hand moves its control only while it does not pinch, and both glide and are centred: a hand at mid height plays what was composed.

24. Shortcuts

```
Space            play / stop
F1               help (this page); Esc closes it
F11              full screen (standalone)
Ctrl+Z           undo
Ctrl+Y           redo (also Ctrl+Shift+Z)
Ctrl+S, Ctrl+O   export the set, load one
Ctrl+E           export MIDI
Ctrl+K           search: any control, preset, page or action
right click      MIDI Learn / Forget MIDI / Default value on any control
double click     a knob back to its default
```

The window can be dragged to any size; the whole panel zooms with it. The standalone's title bar has a maximise button, and Full screen in the settings (or F11) fills the screen.

25. Updates

Phosphene looks for a newer release at most once a day, when its window opens, by asking GitHub for the latest release of the project -- nothing else is sent, and nothing is installed by itself. When there is one, the header says so and a click opens the release page. The check can be switched off here on the help page or in the settings, and Check now asks at once.
